



**SMITH ENGINEERING
HYPERLOOP**
Queen's University
Design Team

Final Design Package for the Not-a-Boring Competition 2026

Vertical Boring Machine

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1. Team Description

Queen's Hyperloop Design Team (QHDT) is a dynamic group of over 150 passionate students from Queen's University in Kingston, Canada. The team shares a vision to change the future of transportation and is committed to pushing the boundaries of innovation and engineering excellence, by leveraging interdisciplinary collaboration amongst the immense talent pool within Queen's University. This submission clearly shows the teamwork and collective passion the team has contributed towards making tunnel boring a more accessible reality.

The motivation for the Tunneling and Boring (T&B) sub team comes from an interest in making working on a multidisciplinary project related to the earth sciences, infrastructure and technology. The team this year will create a management and working structure that will allow junior as well as senior members to contribute. The way this will be achieved is through experimenting with different styles and approaches while diligently recording everything. The foundational team culture will be important to cultivate. The main tenants will be of innovation and collaboration that will sustain itself for years to come.

The T&B team is composed of three different sub teams that work together to ensure the success of the team. The three sub teams are mechanical, drilling, and software and electrical. These teams are each composed of a manager, and general members. The members can be found in the table below.

1.1 Team Members

Table 1 List of team members, with their respective position, discipline, and email contact

Name	Position	Discipline	Email
Conrad Bierman	Co-Captain	Mechanical '25	21cpb10@queensu.ca
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1.2 Team Advisors

Table 2 List of faculty advisors with their current institution

Advisor	Title	Institution
Graham Swarbrick	Tunneling and Boring Advisor	Project Engineer at Herrenknecht AG
Dr. Geoff Eichhorn	Tunneling and Boring Advisor	Civil Engineering Department at Smith Engineering, Queen's University
Nick Bundza	Tunneling and Boring Advisor	Mechanical Design Engineer at Nokia

2. Design Description for Digging Device

The team is looking to design and manufacture a working vertical digging machine prototype for the Not-a-Boring mini competition in Texas 2026. The team has 3 main streams, who have mapped out the machine design summary. The initial site investigation and analysis yielded important information about the geology in the area, which the rest of the design was based around.

2.1 Site Investigation and Design

Important to consider for the design and selection of the preferred tunnelling methods is the geological conditions found in the competition area. The land in the Not-a-Boring competition area has been used for agricultural purposes since 1985. It shows some history of flooding which has altered the depositional environment, introducing gravel layers, sand or silt seams. The primary geological formations in the area include Alluvium containing clay and sand mixed with silt and gravel. These deposits have potential for significant variation over short distances. Terrace deposits are also found in the area, and are deposited by stream beds, and contain, clays, sands, silts and gravels. There is also a Simsboro formation, consisting of mainly coarse-grained sand deposits, overlying mudstones and claystone layers.

The drilling machine only needs to dig one metre deep into the subsurface. Although the design should be applicable to the main competition as well, which requires the TBM to have a minimum distance from the top of their tunnel to the surface. The TBM will likely have to have a system that can deal with soil which is 0.5m and deeper. With this informing the interpretation the boreholes yielded two types of dominant geological materials the team will have to consider for the one-metre-deep hole that the VBM will need to dig. This includes a lean clay with sand and a fat clay. This was provided by the bore hole logs as well as generalized stratigraphy conditions that were provided for each borehole. The generalized conditions provided the basis for this analysis

To be considered within the design are also the site conditions of the area for drilling, Bastrop Texas is located near the Gulf of Mexico and in the competition month of March receives around 53 mm of rain in March and has low humidity the month of the competition. While the temperature tends to range from 24 to

10. This indicates that the site will likely be dry for the competition. This should be considered when performing the soil analysis, as the soil is likely to be quite dry.

2.1.1 Soil Analysis

The geotechnical report offers the team a comprehensive overview of the site conditions in Bastrop, the challenge for the team was to find what parts of the report were relevant to the VBM design. From background research and discussions with advisors, the important soil properties include:

- The soil type and classification
- Density and unit weight
- Plasticity and consistency

These are the most critical items for the team to use for predicting potential machine performance. The Geotechnical report offers several different properties and descriptions of the same soils. For the purpose of the design these were generalized into the analysis shown in Table 1. These values were drawn from the geotechnical report and provide some basic characterization of the site soil profiles.

Table 3 Generalized Soil Material Table

Soil Material	Classification	Borehole Location	Depth	Plastic Limit (PL, %)	Liquid Limit (LL, %)	Density (kg/m ³)
Fat Clay	CH	B-1	0-18'	22	59	1618-1698
Lean Clay	CL	B-2, B-3	0-28', 0-13'	15-16, 12-19	38, 39	1714, 1793

Evidence of the variability which will likely be encountered in the area. A general breakdown of the Site stratigraphy is described below.

B1

- Cohesive layers (Fat Clay and Lean Clay) are the most difficult layers to auger through due to high fines content and plasticity
- Cohesionless layers (sand and gravel), are easier to penetrate, however dense zones indicated by high N will likely slow down auguring process while the loose sand can collapse into the hole

B2

- The cohesive soils are very stiff to stiff making them challenging to auger due to their plasticity and strength
- The cohesionless soils are very dense, thus penetration through them will be difficult, with the possibility of refusal in some layers

B3

- Cohesive soils range widely in strength, some layers are very hard ($N=50/3''$), these layers will be harder to penetrate than the softer clay layers
- The cohesionless soils are loose to medium dense, they have a low plasticity which suggests that they are easier to excavate than the cohesive clays

Since these sites are mainly characterized by lean and fat clays, these types of soils will be investigated in terms of their material properties, based on the geotechnical reports. Primarily, for the purpose of drilling through the material, the types of grain sizes present, along with the shear stress, direct shear, and density of the material.

Boring	Elev	Depth	Classification				LL	PL	PI	Cc	Cu
● B-1		2.0	FAT CLAY (CH)				59	22	37		
▣ B-1		18.0	LEAN CLAY with SAND (CL)				44	17	27		
▲ B-1		23.0								0.72	12.0
★ B-1		33.0	SILTY, CLAYEY SAND with GRAVEL (SC-SM)				16	10	6		
Boring	Depth	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay		
● B-1	2.0	4.75				0.0	8.6		91.4		
▣ B-1	18.0	4.75				0.0	18.0		82.0		
▲ B-1	23.0	75	1.064	0.261	0.088	5.8	87.2		7.0		
★ B-1	33.0	75	3.883	0.338		37.4	44.7		17.9		
Boring	Elev	Depth	Classification				LL	PL	PI	Cc	Cu
▣ B-2		0.0	LEAN CLAY with SAND (CL)				39	16	23		
▣ B-2		6.0	SANDY LEAN CLAY (CL)				40	17	23		
▣ B-2		13.0	LEAN CLAY (CL)				33	15	18		
▣ B-2		18.0	LEAN CLAY (CL)				33	15	18		
Boring	Depth	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay		
▣ B-2	0.0	75				0.6	29.3		70.1		
▣ B-2	6.0	4.75				0.0	37.0		63.0		
Boring	Elev	Depth	Classification				LL	PL	PI	Cc	Cu
● B-3		2.0	LEAN CLAY with SAND (CL)				38	17	21		
▣ B-3		8.0	LEAN CLAY (CL)				49	18	31		
▲ B-3		13.0	SILTY, CLAYEY SAND (SC-SM)				19	14	5		
★ B-3		28.0	WELL-GRADED SAND with GRAVEL (SW)							1.11	14.3
Boring	Depth	D100	D60	D30	D10	%Gravel	%Sand	%Silt	%Clay		
● B-3	2.0	4.75				0.0	18.2		81.8		
▣ B-3	8.0	4.75				0.0	0.3		99.7		

Figure 1 Grain size distribution figure.

The grain size distribution figures indicate that, for site B-1, which is predominantly fat clay at the depth in question, 100% of the material passes through the 4.75 mm sieve. This means the largest soil type is sand, at 8.6%, while the remaining portion is clay. This suggests there may be minimal irregularities within the soil, such as large sand grains, which could potentially jam an auger turning within a conduit. It could also be inferred that a cutting tool optimized for clay would be best suited for this site. When investigating the other borehole data, B-2 and B-3, the sand constitution of the soil increases. In borehole B-2, for example, sand constitutes about 29.3-37% in the depth the competition is concerned with, which is a contrast compared to the fat clay soil profile. These sites would benefit less from a cutting tool that is mainly optimized for clay, and jamming between mechanical moving parts becomes more of an imminent problem. These factors need to be considered when procuring a design for the vertical boring machine.

The density of the material is also important to consider, as the more the tool progresses into the soil, lateral Earth pressure becomes important. This is because with greater depth, more soil material creates pressure against the submerged tunnel boring machine, especially at the soil-shield interface, where deformations may occur. Knowing the exact densities of the soil would enable more accurate design calculations to predict the loads on the submerged shield, and experiment with various bracing systems which would provide some resistance to deformation or distribute it more effectively. From these reports, the densities of the materials range from 1618-1793 kg/m³.

Direct shear is also an important consideration, since standardized tests from site materials would provide a maximum stress value in which the soil types require to displace material. Again, this is important when designing a drilling or cutting tool along with the required torque to create the required force to excavate material. These tests indicate a shear stress, normal force, and displacement value. The shear stress is the horizontal stress, the normal force the force from above, and the displacement being how much the soil has moved because of the forces applied. From the figure below, 90,000 – 34,500 N/m² of shear stress are required for normal stresses of 160,000 – 20,000 N/m², which is roughly between 1-9 m of soil.

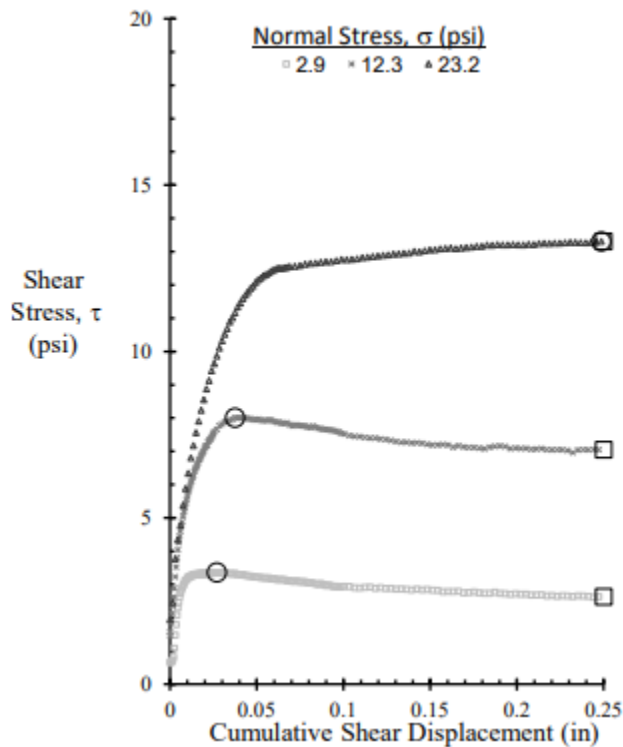


Figure 2 Shear stress graphs B-1

The figure for site B-3 follows below. These involve 124,000 – 24,000 N/m² of shear stress under 160,000 – 20,000 N/m² of normal stress. Site B-3 is mostly a lean clay, so this chart may also be used to describe site B-2.

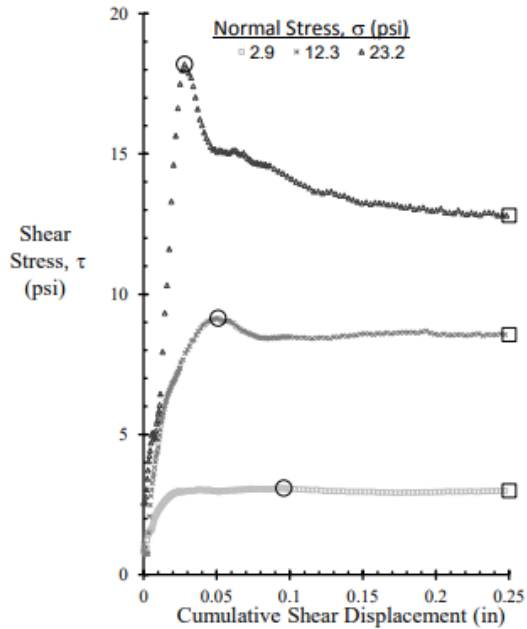


Figure 3 Shear stress graphs site B-3

The stress-strain data for the soil can also provide useful insights. For example, the yield strength associated with the strain involved. For example, for site B-1, shown in the figure below, a yield stress of 1.2 MPa is associated with a 4.8% strain (ex. a 4.8mm deformation over an original 100mm). This is the resistance a tool would need to overcome to product displacement required for excavation. The charts for sites B-2 and B-3 follow. Unlike steel, reinforced concrete, and wood, soil is not purely elastic and does not return to a pre-ultimate yield strength state when stress is released.

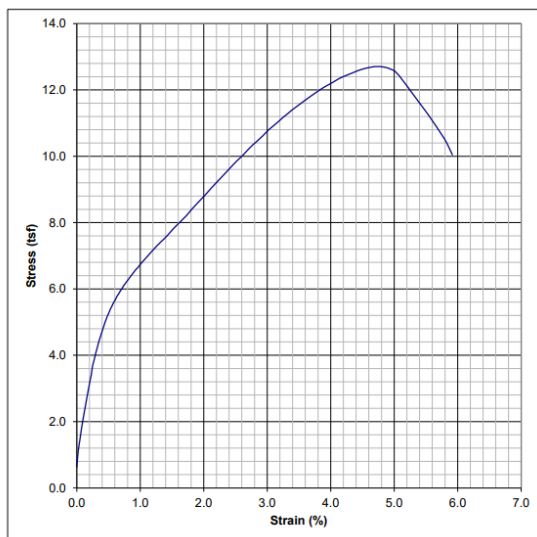


Figure 4 Stress strain curve B-1

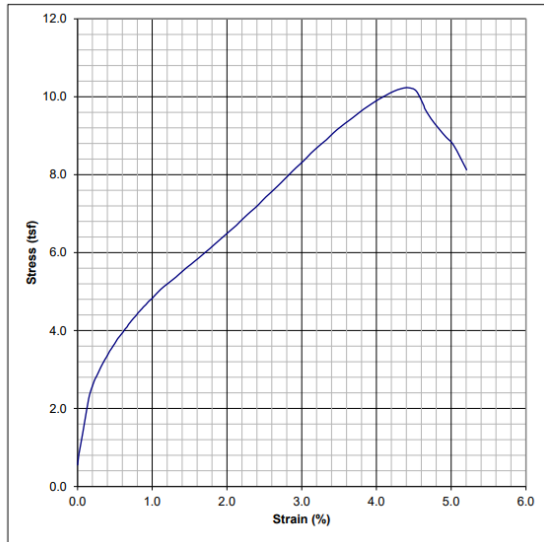


Figure 5 Stress strain curve B-3

2.1.2 Soil Testing Design

To accurately test the tunnel boring machine prior to competition, it would be ideal to run the machine in a setting which simulates the three borehole log sites. This can be accomplished by investigating the geotechnical report data and attempting to find similarly characterized areas local to Kingston, Ontario to test the tunnel boring machine. Alternatively, using the geotechnical reports and the soil grading description, these soils can be replicated using the appropriate aggregate sizes through a soil grading process. Using the information from the geotechnical reports as a guideline, the competition site soil would be recreated in a lab setting and then transported to a testing site. This site would be an excavated hole 1 m deep and 0.5 m in diameter. The hole would be filled with the site-specific designed soil, compacted, and watered (to obtain a similar moisture content). The figure, previously shown, outlines specific depths with specific soil percents for certain sites. For example, in site B-1, it details 8.6% sand and 91.4% clay/silt. As a 1-kg mix, this would constitute 914 g of clay/silt and 86 g of sand. This would be scaled up to the required volume to fill the test hole.

Once the test hole is prepared, the vertical tunnel boring machine would be set in place and run. Two metrics could be proposed to measure performance:

Vertical displacement:

A vertical measurement could be taken at a reference position and checked every 10 seconds to track the VBM's progression into the soil. These measurements could be plotted on a x-y scatterplot as a function of time for further analysis. The recording intervals could be changed after initial results as more ideal metrics are discovered.

Mass displacement:

A mass measurement can be taken from the refuse auger moving material from the base of the boring machine to the surface. This mass can be cumulatively measured every 10 seconds and plotted on an x-y

scatterplot as a function of time for further analysis. As with the vertical displacement method, recording intervals or parameters can be refined as more ideal metrics are discovered.

2.2 Machine Top-Level Design Summary and Layout

2.2.1 Cutter Head

The cutter head has been designed this year to improve excavation efficiency in the soft, fine-grained soils identified during the site investigation, which revealed a high fines content of 70–91%. The most significant design change is the fundamental shape of the face. There was an increase in the open ratio to 67%. This ratio is significantly different from the more conservative 40%, which is typically used in harder ground conditions. This adjustment is ideal for the vertical boring design as gravity naturally contains excavated soil, preventing any material from moving back through the cutter face.

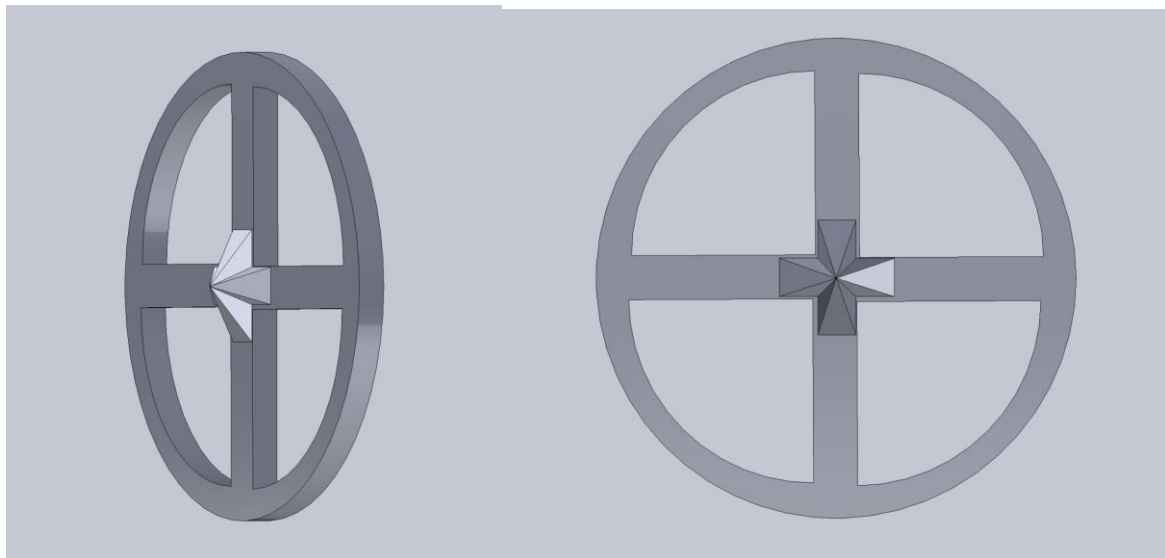


Figure 6 Cutter head design with new opening ratio of 67%. This excludes the teeth that will be attached to the face.

With that greater opening area, the design can focus on maximizing excavation throughout. A larger open area will give potential for a greater volume of soil to move into the chamber during each rotation. This will directly improve the machine's advance rate. The geometry of the four openings is radial slots, with their curved bases. This shape was chosen to specifically draw material away from the face and guide it efficiently toward the screw conveyor, promoting smooth flow and decreasing the likelihood of clogging.

The flat face layout, identified as optimal for this soft geology, incorporates cutter teeth that remove material effectively with minimal energy demand, as well as allowing for bidirectional rotation. The entire cutter head is machined from a single high-grade steel piece. This will provide the rigidity and durability necessary for demanding excavation conditions.

2.2.1.1 Cutter Teeth

The cutter head uses a bidirectional “beaver tooth” cutter arrangement that allows efficient excavation in both forward and reverse direction. The functionality of the teeth will help improve the digging method, as changes in direction allow for continuous soil excavation. The head incorporates 16 teeth along with a central

scraper, placed to perform reliably in sticky clay conditions. The teeth are positioned in tightly spaced groups of four on each wing of the cutter head, eliminating open gaps in the central zone and reducing the probability of clogging.

The central scraper begins the cutting action by breaking the soil and directing it toward the openings, while the surrounding teeth carry out most of the excavation. All cutting materials will be welded to the face to ensure secure attachment and integrity during high operational loads.

2.2.2 Cutterhead Rotation System

The cutterhead will be powered by a dedicated motor in the mechanical body, ensuring sufficient torque is available for the excavation of dense, saturated clay. This motor will connect to a rigid drive shaft that transfers rotational energy down the length of the tunnel boring machine to the cutterhead assembly. To accommodate potential misalignments and handle significant axial and radial loads generated during digging, the drive shaft will be supported by high-capacity ball bearings. This robust mechanical system is designed to provide the reliable, high-torque rotation necessary to fracture the soil and impart the centrifugal force required to move the disturbed material toward the auger intakes at the shield's periphery.

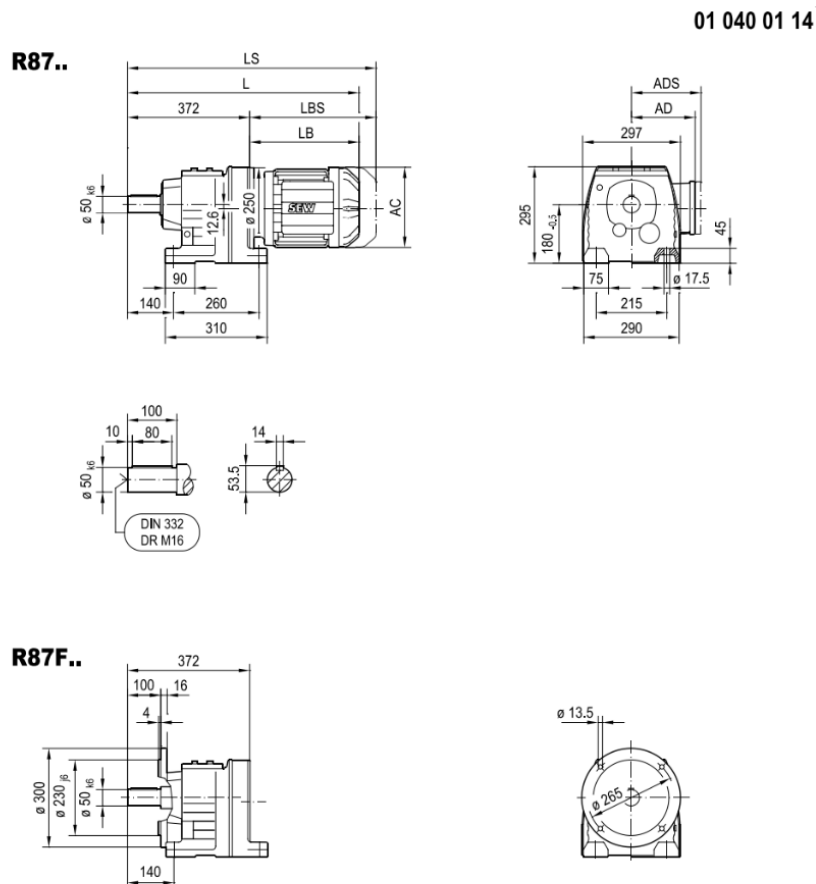


Figure 7 Cutterhead rotation motor specifications

2.2.3 Soil Removal System

The excavated soil needs to be removed from the shield after digging has been done by the cutter head. The material will come through the cutterhead and due to the rotating nature of the cutterhead will collect on the outsides of the shield. The auger chamber will be positioned to collect soil at the outer rims of the cutterhead and then be carried upwards by an auger. The auger will be around 0.01m in diameter with an enclosure surrounding it.

The removal system will be provided power by a dedicated motor. The disturbed soil will build up in the bottom chamber of the of the shield, gaining upwards pressure from the continuous adding of soil in the duration of the dig. The spin of the cutterhead directs the clay to the outer diameter inside the shield, guiding it to the auger screw. There is also a catch present at the bottom of the auger shield to centralize the soil in the diameter of the auger. Once the soil is centralized, the rotation of the auger transports the soil upwards. The plan for soil disposal once the transported soil reaches above surface level is to add some sort of slide so that it can be relocated far from the hole so that it will not spill back into what has already been dug.

2.2.4 Soil Removal Rotation System

The auger will be powered by a separate, independent motor, also mounted on the dynamic platform. This design allows for the precise control of the auger's operational speed, which has been specified at 170 RPM, independent of the cutterhead's rotation. This is critical for optimizing the soil evacuation rate to match the cutterhead's production. The motor will connect directly to the auger shaft, which is housed within its own enclosure. The use of a dedicated motor simplifies the control scheme, enabling the team to fine-tune transport efficiency and quickly respond to any clogging by adjusting the auger's RPM without affecting the primary digging process.

2.2.5 Shield and Support

The VBM's shield will be constructed from High Density Polyethylene due to its high strength-to-density ratio, allowing ample protection of the internals of the machine. The shield will be supported by an endoskeleton-like structure on the interior, which will give it far more strength as well as allow the motor to be mounted easier. This internal structure will be inside all 3 pieces that will make up the VBM. The endoskeleton-like structure will be constructed from Carbon Steel and will be machined into disks for the VBM and will be reinforced by Carbon Steel beams on the inside, as shown in the figure below, displaying the middle of the 3 pieces of the VBM.

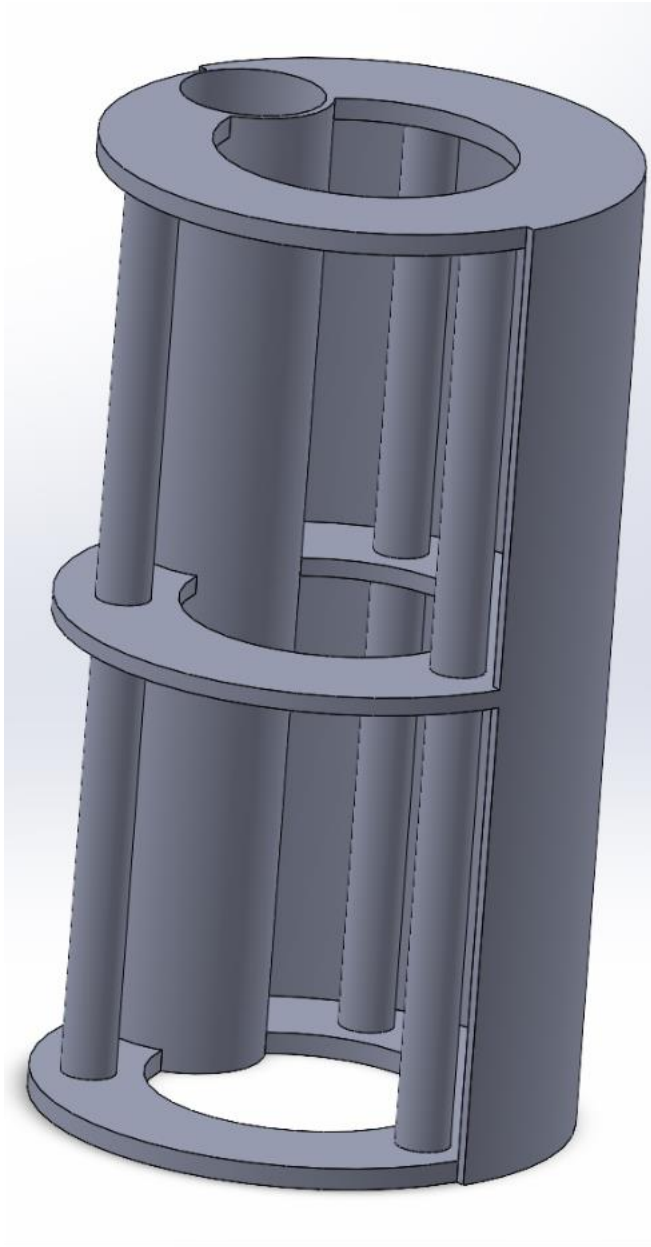


Figure 8 Internal structure of the VBM

SolidWorks FEA analysis was performed on the VBM's shield, using HDPE as the material. 130kPa was used as the estimated maximum force that could be put onto the VBM while digging, which was used in the analysis. Since the load from the soil is distributed over such a large area, as well as the force should be much lower than 130kpa, there is minimal stress on the shield. It is practically impossible that the force of the soil pressing onto HDPE could compromise the shield, and it will likely be far more durable than what is necessary. The internal structure helps reinforce the shield even further.

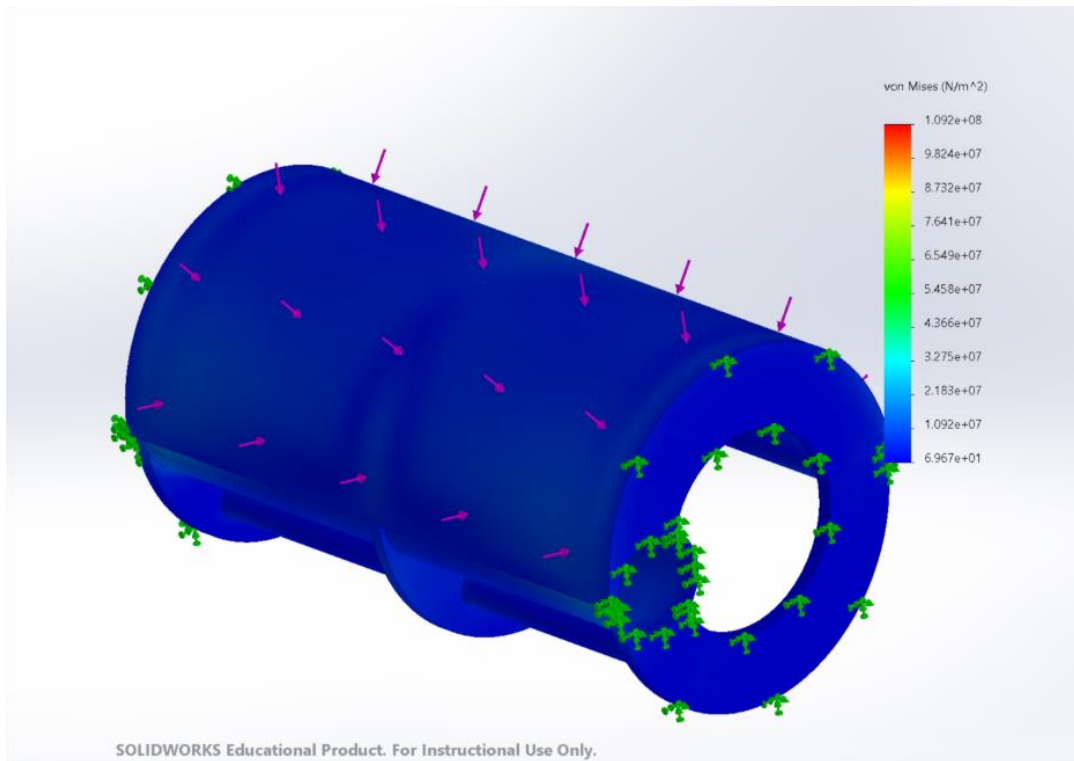


Figure 9 FEA Analysis on the internal structure

2.2.6 Protection Around the Machine

The frame itself will serve as one of the main protections around the machine. It will ensure that it maintains a stable excavation path and ensuring that it isn't damaged from the ground. The machine itself will have casing around the main body, and an additional protection around the auger itself. This will ensure that any material excavated in the auger remains safe, and nothing gets into the machine itself. The team will assess the overall success of these casings and may add additional protection around the motor.

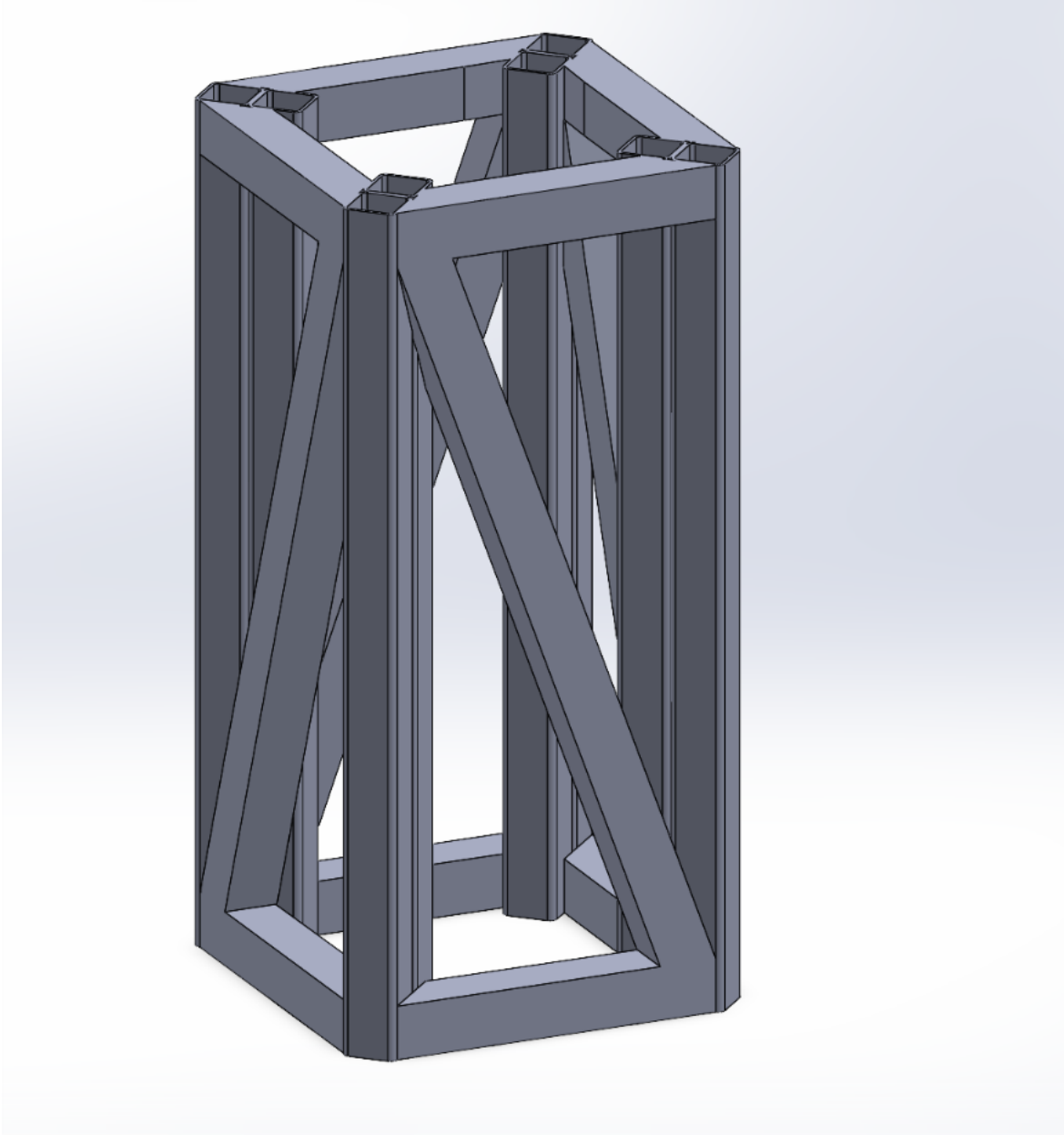


Figure 10 Frame design of the VBM

2.2.7 Drill Lift System

Currently, no lift system is being considered for the design. The VBM will be secured to the frame via a rail system on each corner of the frame, allowing the VBM to lock into place and slide freely. An example of this is shown below.



Figure 11 Example of guide rail for the lifting system

For driving force downwards, the weight of the machine will provide around 3.924kN of force which is calculated using an estimate of the VBM's weight of ~800kg. For lifting the machine, an attachment will be added to the top of the VBM to allow a forklift to remove it from the boring site. The load of this lifting is analyzed in section 5.4.

2.2.8 Power System

The VBM power distribution system utilizes a split-voltage architecture designed to isolate sensitive control logic from high-voltage motor noise while ensuring fail-safe operation. The system draws power from the competition-provided 100kW generator via two distinct feeds: a 480 VAC 3-Phase line for the main cutter drive and a 120 VAC 1-Phase line for the auger and actuation subsystems.

High Voltage (480 VAC): The primary 480V feed powers the 12.3 HP Cutter Motor. This circuit is protected by a 60A fused disconnect and a forced-guided Safety Contactor. A Variable Frequency Drive (VFD) manages the motor's inrush current and allows for precise torque and speed control.

Low Voltage (120 VAC & 24VDC): The 120V feed powers the 0.25 HP Auger Motor and the 24VDC Control Actuation bus. The Auger motor is controlled via a reversing contactor pair interlocked for jam-clearing operations. A DIN-rail power supply converts 120V AC to 24V DC to energize the safety relay logic, contactor coils, and stack lights. This ensures that a loss of facility power immediately de-energizes all actuators ("De-energize to Trip").

Isolated Control Power: To eliminate ground loops and prevent motor noise from disrupting the microcontroller, the Control Logic (Raspberry Pi 5 and Teensy 4.1) is powered independently by a stored-energy system (USB-C Power Bank). This galvanic isolation ensures the main control of the machine remains active to log data and communicate status even if the main power is severed.

2.2.9 Software System

The VBM software architecture utilizes a hierarchical Dual-Controller Design to ensure separation between high-level user interface tasks and real-time safety-critical control. This split architecture ensures that the machine remains responsive and safe even in the event of a graphical interface freeze or telemetry lag.

The system consists of two primary computational nodes:

1. **The Host Computer (Raspberry Pi 5):** It runs the Operator GUI (developed in Python/Tkinter), manages the SQLite data logging database, and handles high-bandwidth telemetry broadcasting. It acts as the command center, sending high-level directives (e.g., "Start Digging," "Retract") to the lower-level controller.
2. **The Safety Microcontroller (Teensy 4.1):** Responsible for "Real-Time Control." This MCU executes the core safety state machine, directly reads sensors via the ADC, and controls the MOSFET drivers for the motor contactors. It operates on a deterministic loop to ensure precise timing for sensor polling and watchdog's "heartbeat" generation.

Communication between these nodes is established via a primary Ethernet link using a ZeroMQ (ØMQ) Publish-Subscribe protocol, ensuring low-latency data exchange. A secondary UART serial link serves as a hardware fallback. This architecture allows for robust "De-energize to Trip" safety logic; if the Safety MCU stops receiving a "Heartbeat" signal from the Host Computer, it autonomously executes a safe shutdown sequence, de-energizing all actuators to prevent uncontrolled operation.

2.2.10 Digging Machine Parameters

The maximum advance rate that the machine will exhibit is $4.5 * 10^{-4}$ m/s. This is calculated using a penetration rate of 1.8mm/rev, which was selected by the team. The RPM will be 15 for the cutterhead system, which was also incorporated into the calculation. These calculations can be further seen in 3.2.1. Estimate of Mechanism Throughput The time that is estimated is 7.4 minutes.

$$Time = \frac{Material\ Excavated}{Advance\ Rate}$$

$$Time = \frac{0.2\ m^3}{4.5 * 10^{-4} * 60\ \frac{m^3}{min}}$$

$$Time = 7.4\ mins$$

2.2.11 Machine Dimensions and Weight

Table 4 Dimensions and weight of major components

Component	Dimensions (L x W x H)	Mass (kg)	Notes
Main Electronics Enclosure	600 mm x 600 mm x 300 mm	20.0 kg	NEMA-rated steel enclosure housing safety logic, contactors, and low-voltage distribution.
Variable Frequency Drive (VFD)	250 mm x 150 mm x 180 mm	8.0 kg	Delta VFD220E43A (Mounted inside or adjacent to main enclosure).
Power Cabling (480V)	30 m Length (100 ft)	30.0 kg	8/4 SOOW heavy-duty copper cable for Cutter Motor feed.
Power Cabling (120V)	30 m Length (100 ft)	7.0 kg	10/3 SOOW cable for Auger Motor and Control Power feed.
Cutter Face	0.5 m Diameter	12.75 kg	The total volume is 1624.82 in centimeters.
Auger	4" Diameter Auger, 1.6m length	0.95kg	This weight includes both auger pieces that will be needed combined
12.3 HP Motor	1m length, 13"x12" body	136.0kg	VBM's Primary Motor for Cutterhead motion
VBM Top Piece	0.5m diameter, 30cm height	32.0kg	The Top Piece of the VBM, housing electrical components and more.
VBM Middle Piece	0.5m diameter, 1.1m height	82.0kg	The Middle Piece of the VBM, housing the motor and gearbox.
VBM Bottom Piece	0.5m diameter, 20cm height	28.0kg	The Bottom Piece of the VBM, housing the soil containment area, bottom of auger, and cutterhead attachment.

3. Machine Design Descriptions

3.1. Machine Thrusting Mechanism

The first section of calculations showcases the calculations and assumptions that were made to calculate the power, and torque that was required by the main body shaft of the machine. This section will be operating

the cutterhead. Several properties were taken from the geotechnical report to represent the soil that was being worked with. To ensure accuracy and room for error within the design, the highest stress values were taken from the report. The team wanted to ensure that regardless of the soil conditions we would be supplying the machine with enough power and torque. One of the main concerns that we were confronted with was the machine not being able to dig through the soil due to underestimating in preliminary torque and power calculations.

$$c = 0.9psi$$

$$c = 6205 Pa$$

$$\phi = 33^\circ$$

$$\sigma = 26.56 psi$$

$$\sigma = 183,231 Pa$$

$$u = \lambda_w h$$

$$u = 9810 Pa$$

$$\sigma_e = \sigma - u$$

$$\sigma_e = 173,314 Pa$$

$$s = c + \sigma_e \tan \phi$$

$$s = 6205 + 173314 \tan(33)$$

$$s = 118,756 Pa$$

The shear strength of the soil was calculated using different soil properties. The effective stress was calculated by subtracting the stress that was likely being caused by water. This was used alongside cohesion and friction angle to calculate the shear strength of the soil itself. As mentioned above, this calculation is on the highest side of soil stress that could exhibited on the competition property.

The first force that was calculated was the force of the cutterhead face within the soil. This was calculated using the shear strength and the area of the cutterhead. The opening ratio that was selected for the cutterhead was 0.67. This was selected based on research on cutterhead faces. The torque of this force was also calculated.

$$F = SA$$

$$F = 118,756 \frac{\pi}{4} (0.5^2) (1 - 0.67)$$

$$F = 8553 \text{ N}$$

$$T = Fr$$

$$T = 2140 \text{ N m}$$

The next force that was calculated was the force of each individual tooth digging. The primary purpose of the teeth is to agitate the soil. The friction force of the soil must be considered in these force calculations.

$$F_{Tooth} = \sigma \mu A$$

$$F_{Tooth} = 200 \text{ N}$$

$$F_{Tooth, Total} = N \cdot F_{Tooth}$$

$$F_{Tooth, Total} = 3200 \text{ N}$$

$$T_{Tooth} = FR$$

$$T_{Tooth} = 1200 \cdot (0.25)$$

$$T_{Tooth} = 800 \text{ N} \cdot \text{m}$$

The last force that was incorporated into the calculations was the friction of the face of the cutterhead. The face will still experience friction against it.

$$F_{Friction, Face} = \sigma \mu A$$

$$F_{Friction, Face} = 183321 \cdot 0.5 \cdot \frac{\pi}{4} (0.5)^2 (1 - 0.67)$$

$$F_{Friction, Face} = 5,939 \text{ N}$$

$$T_{Friction, Face} = 5939 \cdot 0.25$$

$$T_{Friction, Face} = 1485 \text{ N} \cdot \text{m}$$

$$T_{total} = 4500 \text{ N} \cdot \text{m}$$

The speed of 15 RPM was selected for the cutterhead. This was on the higher end of the range we deemed acceptable. The speed was also a limiting factor for our motor, and for the complexity of the gearbox design. Choosing a higher speed allowed us to simplify the speed reduction, while remaining successful.

$$P_{cutterhead} = 4.5 \cdot 15 \cdot \frac{2\pi}{60}$$

$$P_{cutterhead} = 7 \text{ kW}$$

$$P_{cutterhead} = 10 \text{ hp}$$

To include a factor of safety and understanding that the machine will likely need additional power to be successful in the competition, the motor that will be selected is 12.3 hp. This gives the motor to have an efficiency of 80%.

3.1.1. Axial Friction Analysis

The main frictional force that was calculated was the friction that is against the face of the cutterhead. Due to the nature of the design, the face will be rubbing against the soil and creating friction. While the exact coefficient of friction of steel against the clay material wasn't provided, the value was assumed to be 0.4. This was estimated based on research of similar situations. The second friction force that must be considered is the friction against the face of the tooth. The tooth is agitating the soil to ensure that it can be excavated. The total axial friction is 12,000 N or 12 kN. These friction forces make up the majority of the force calculations of the cutterhead.

3.1.2. Torque Friction Analysis

The torque friction analysis uses the axial friction forces that were calculated and multiplied by the lever hand of the force. For this instance, it was the radius of the cutterhead. The full calculations can be seen in the calculation section above, but the total torque from friction of the cutterhead is 3110 Nm, which makes up approximately half of the torque on the cutterhead.

The remaining force is supplied by the pressure and pushing force of the cutterhead face. This force supplied the remaining 3890 Nm of torque. This was calculated using the properties within the soil (cohesion and stress) to calculate the shear force, and subsequently the force on the cutterhead face. The total torque from the cutterhead was 7000 Nm.

3.2. Muck Removal System

3.2.1. Estimate of Mechanism Throughput

To understand the mechanism throughput and how much material and power was needed for the system, the first step that was taken was selecting a penetration rate. From this the advance rate was calculated, which allowed the team to gain a better understanding of the mass flow rates. The penetration rate was selected at 1.8mm/rev.

$$p_{th} = 1.8 \frac{mm}{rev}$$

$$v_{th} = p_{th} \cdot \eta$$

$$v_{th} = 1.8 \cdot 15$$

$$v_{th} = 27 \frac{mm}{min}$$

$$v_{th} = 4.5 \cdot 10^{-4} \frac{m}{s}$$

The advance rate was used to calculate the flow rate, which was done using the advance rate. Multiplying it by the area allowed the team to have a quantitative value of the flow rate.

$$Q = A \cdot v_{th}$$

$$Q = 0.196 \cdot (4.5 \cdot 10^{-4})$$

$$Q = 8.82 \cdot 10^{-5} \frac{m^3}{s}$$

$$V_{rev} = n \left(\frac{\pi}{4} \right) (D^2 - d^2) P$$

$$V_{rev} = (0.67) \left(\frac{\pi}{4} \right) (0.1^2 - 0.008^2) (0.005)$$

$$V_{rev} = 3.12 \cdot 10^{-5} \frac{m^3}{rev}$$

The RPM of the auger was calculated through multiplying the flow rate by 60 to convert to min and then dividing by the volume per revolution.

$$\eta_a = \frac{60 \cdot Q}{V_{rev}}$$

$$\eta_a = \frac{60 \cdot (8.82 \cdot 10^{-5})}{(3.12 \cdot 10^{-5})}$$

$$\eta_a = 170 \text{ RPM}$$

Using the selected RPM, the team was able to calculate the flow rate of soil going through the water. The soil will be wet, so the wet density was calculated using the percentage of water that was within the soil. This value was used to calculate the mass flow rate of soil traveling throughout the auger.

$$Q_A = \frac{(V_{rev} \cdot \eta_a)}{60}$$

$$Q_A = \frac{(3.12 \cdot 10^{-5}) \cdot 170}{60}$$

$$Q_A = 8.8 \cdot 10^{-5} \frac{m^3}{s}$$

$$\rho = \rho_d(1 + w)$$

$$\rho = 1800(1 + 0.17)$$

$$\rho = 2106 \frac{kg}{m^3}$$

$$\dot{m} = \rho Q_a$$

$$\dot{m} = 2106(0.88 \cdot 10^{-4})$$

$$\dot{m} = 0.186 \frac{kg}{s}$$

The mass flow rate was multiplied by the gravitational acceleration constant and the distance that the material would have to travel to determine the power required by the system. The team has assumed that there will be some general inefficiencies within the auger, and that additional power should be accounted for. The final power requirement of the auger system will be 0.25 hp.

$$P_{Lift} = \dot{m}gH$$

$$P_{Lift} = 29 \text{ W}$$

$$P_{shaft} = \frac{29}{0.15}$$

$$P_{shaft} = 193 \text{ W}$$

$$P_{shaft} = 0.25 \text{ hp}$$

The final step in this process was to calculate the angular frequency, which was just done by converting the speed from RPM to rad/s. Then the torque can be calculated by dividing the power by the angular frequency.

$$\omega = \frac{2\pi n}{60}$$

$$\omega = 17.8 \frac{rad}{s}$$

$$T = \frac{P}{\omega}$$

$$T = \frac{193}{17.8}$$

$$T = 10.8 \text{ N} \cdot \text{m}$$

3.2.2 Muck Transport System

The transport system of the muck will be an auger that is enclosed within a case of HDP plastic to ensure that the waste remains enclosed. The augers will bring the material to the surface, where it will then be discarded. The exact design of the removal system is not fully done, but the team has conceptually thought of the requirements. At the end of the auger there will be a chute that will allow the material to exit the machine and be discarded beside the machine. This is necessary as the auger is much narrower than the hole that is being excavated and handles a lot more material per revolution than the cutterhead.

3.3. Machine Structural Analysis

3.3.1 Digging Machine

For the digging machine analysis, output torque calculations from 3.1 as well as output torque listed on the SEW EURODRIVE website for a 12.3 HP, 54rpm helical gearmotor, which is what will most likely be purchased for our design, will be used. The direct output torque is 1.7kNm, which is taken from the technical data of the helical gearmotor on the SEW EURODRIVE website. After the gearbox, the force which was calculated in 3.1 is 7kN. Below is the design setup that will be used inside of the VBM, using the combination of the helical gearmotor's ratio and a simple gearbox to reduce the rpm.

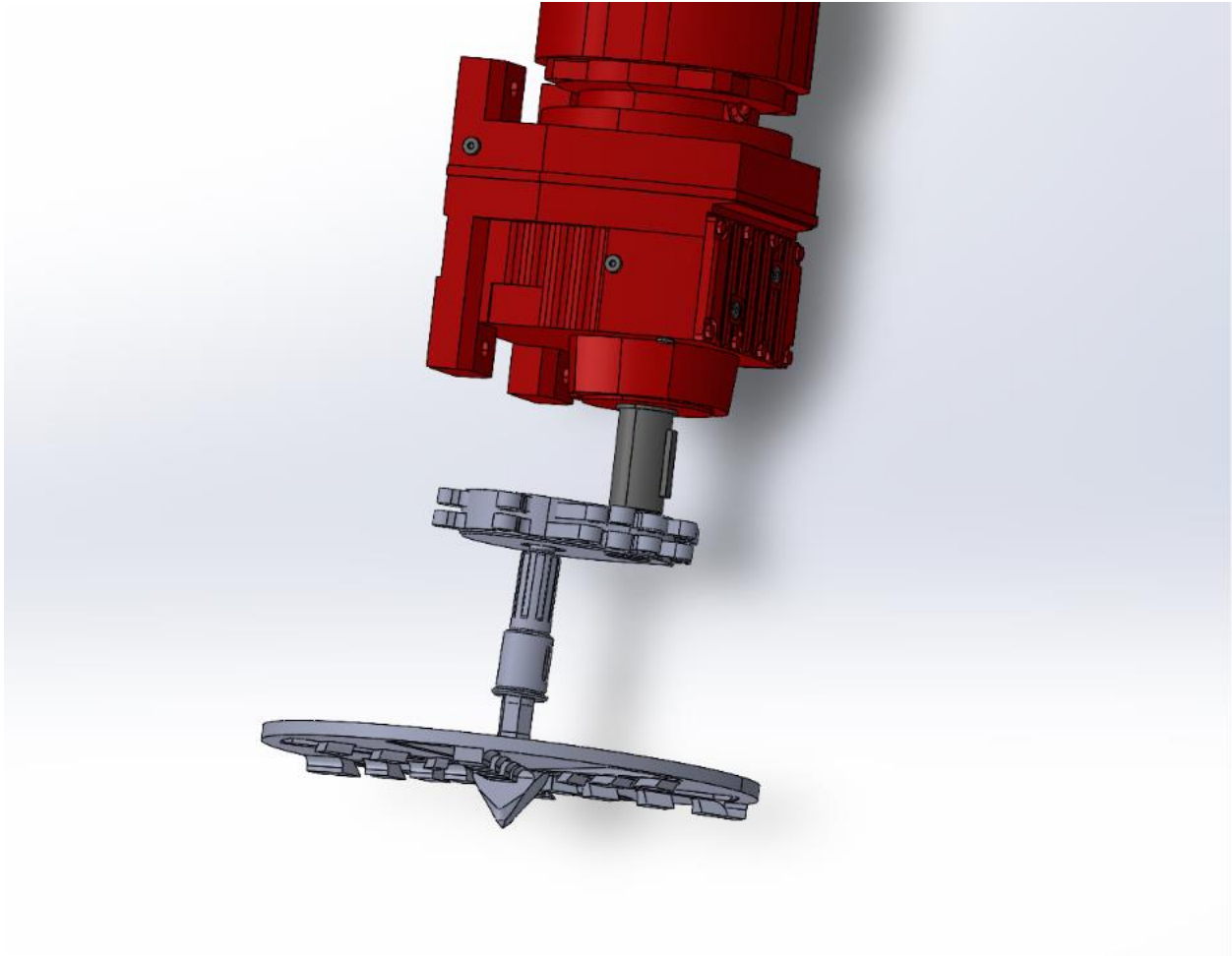


Figure 12 Cutterhead motor, and shaft to the gearbox and cutterhead

FEA analysis was done on the shaft connecting the gearbox to the cutterhead, as the gearmotor will either directly connect into the gearbox or have a very short and strong shaft. A model of a very similar shaft to what will be used was used in this analysis, as a specific shaft has not been chosen yet due to the final gearbox design being in progress. It is similar shape as well as material, being cold drawn steel. The shaft will need to be chosen carefully to be able to sustain the 7kNm of force that will be exerted on it.

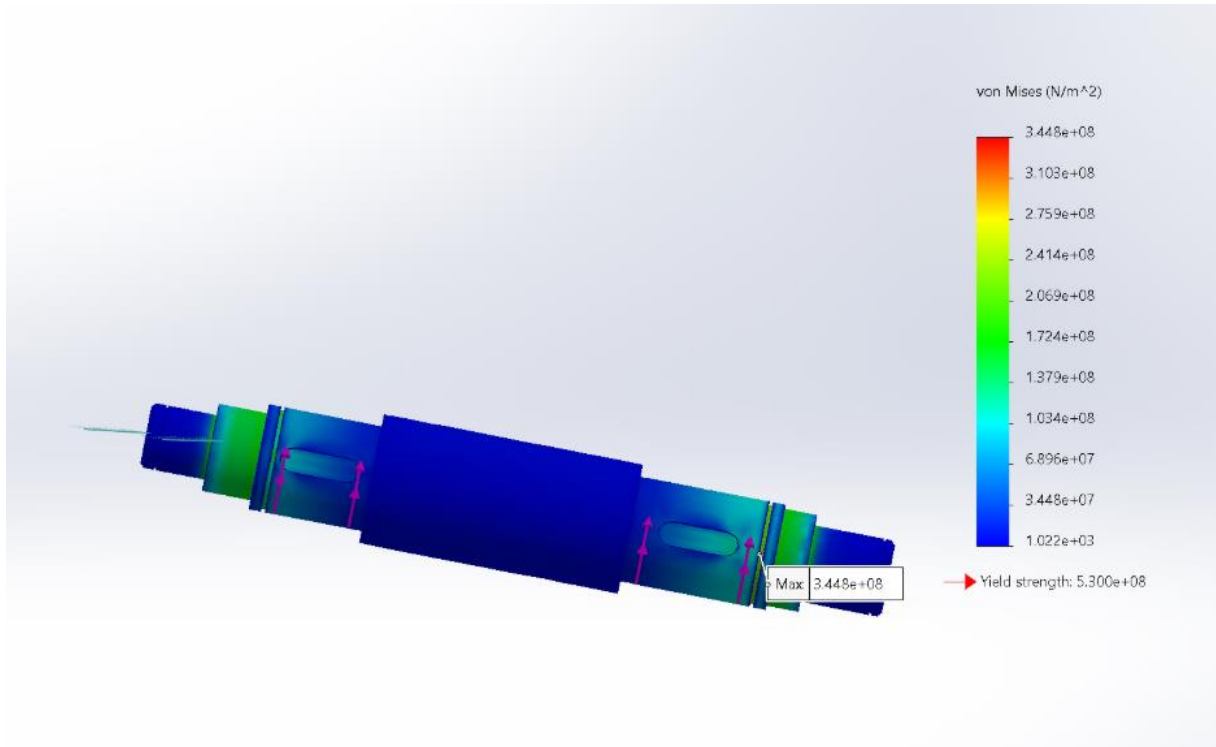


Figure 13 FEA analysis on cutterhead shaft

As shown in the analysis, the brunt of the strain will be on where the shaft connects to the cutterhead. Maximum torque will be near the ends of the shaft. It will be noted when finalizing the attachment that it will need to be reinforced with welds and screw connections to allow for ample resistance against the torque of the gearmotor. In this analysis of a similar shaft, it was able to handle the strain of the motor and did not exceed the yield strength, though this analysis is an overestimation of the force due to the cutterhead being static in this case. Since the cutterhead will cut through the soil, or slip, the force on the shaft should be far lower. A strong shaft will still be chosen in case the cutterhead encounters resistance, which is when high torque would be applied to the shaft.

Failure Mode can be defined as any point in the system where the yield strength was exceeded by the stress applied and would cause permanent inelastic deformation. As mentioned before, this analysis was an overestimate, though the shaft should be chosen to be able to withstand high strain due to it being possible that a large amount can be applied to it when digging. The Factor of Safety (FoS) that should be aimed to achieved is 1.5 and will be calculated using the below formula.

$$FoS = \frac{\sigma_{y,vM}}{\sigma_{vM}} = \frac{(5.3)10^8}{(3.44)10^8} = 1.54$$

The Factor of Safety for this torsion load being applied is 1.54, which barely exceeds the aimed value. A stronger shaft will likely be chosen, but one extremely similar to this would work, as it would theoretically not

fail, even if the cutterhead ran into resistance. When a shaft is finalized and purchased, it will be chosen with enough strength to achieve this 1.5+ Factor of Safety.

3.3.2. Launch/Supporting Structures

Our frame will be designed using Hollow Structural Steel that was received last year and will be subjected to torque from the rotation of the cutterhead. By holding the machine in place, it will support and resist these forces, allowing the machine to more effectively bore. SolidWorks FEA analysis was used to model this force and allowing a deeper understanding of where additional reinforcement may be added to the frame. The force on the frame can be calculated based off the cutterhead torque and divided by the 4 corner beams. The moment generated by the VBM is around 7kNm, so each corner should receive 1.75kN of torque. Since the frame is symmetrical, it can sustain either clockwise or counterclockwise torque. As shown in the analysis below, this torque is applied via a perpendicular force on each beam.

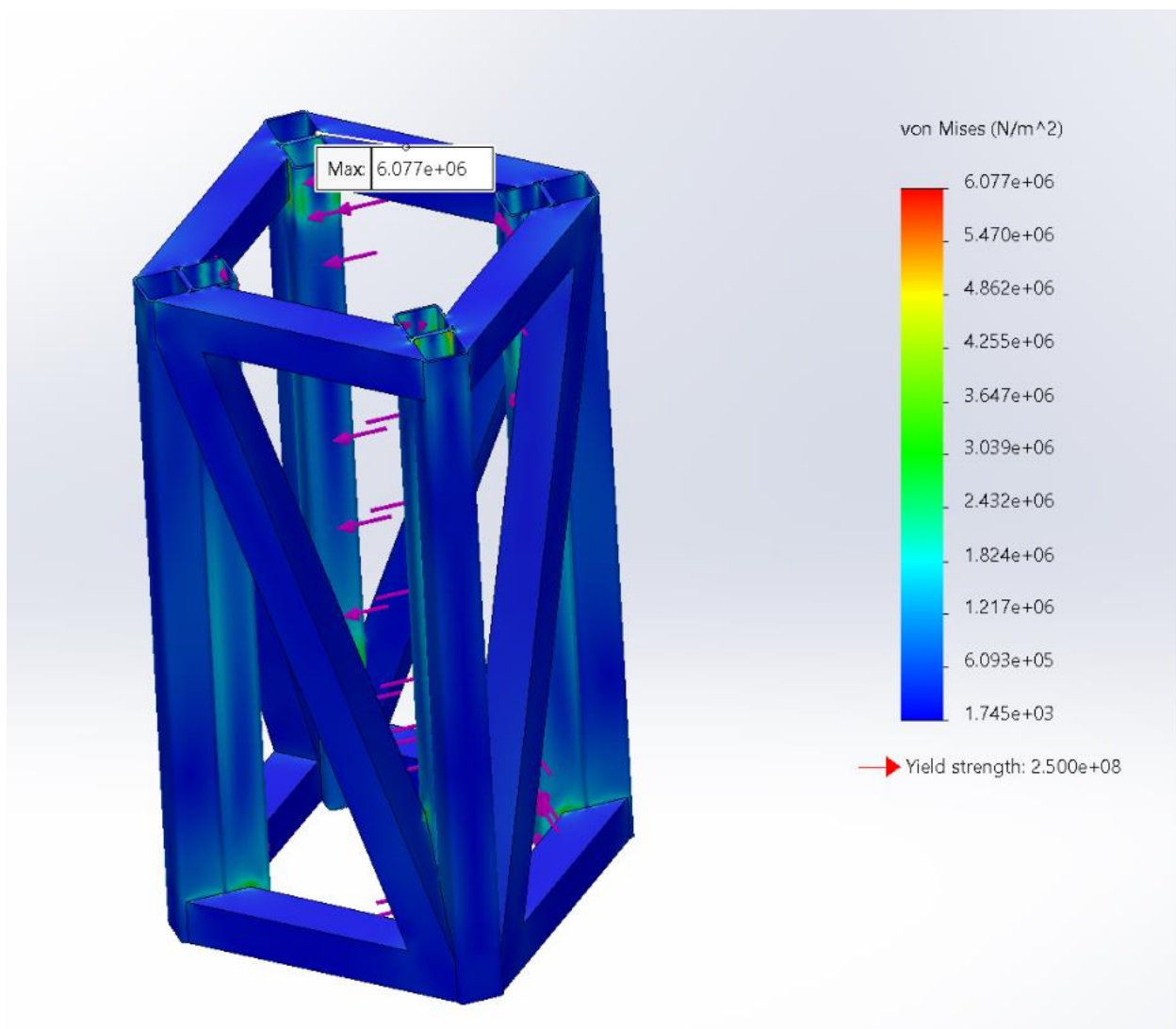


Figure 14 FEA analysis on VBM frame

Similar to the internal digging machine, Failure Mode can be defined as any point in the system where the yield strength was exceeded by the stress applied and would cause permanent inelastic deformation. In this case, the frame should not even come close to being deformed, and the below calculation for Factor of Safety supports this.

$$FoS = \frac{\sigma_{y,vM}}{\sigma_{vM}} = \frac{(2.5)10^8}{(6.077)10^6} = 41.14$$

As mentioned in 3.3.1, the FoS that is aimed to achieve is at least 1.5, and the HSS frame of the VBM has a FoS of 41.14, which is much higher than what is required. It was made to be much stronger so that it is heavier, to allow for more friction on the soil to resist the torque of the VBM. No additional reinforcement is required to add to the current design; the frame will be welded/bolted together.

3.4 Electronics System

3.4.1 Sensor List and Locations

The electronics system is architected into three primary subsystems to ensure safe operation and precise control: the Control Logic System, the Power & Actuation System, and the Telemetry & Feedback System.

The Control Logic System serves as the central processing unit for the machine. It consists of a Raspberry Pi 5 (Host PC) for high-level GUI and logging operations, and a Teensy 4.1 microcontroller (Safety MCU) for real-time control. An ADS1115 Analog-to-Digital Converter (ADC) is utilized to interface 5V analog sensors with the 3.3V logic of the Teensy. These components are housed in a vibration-dampened electronics enclosure located at the top of the machine to minimize exposure to debris and mechanical shock.

The Power & Actuation System manages the high-voltage distribution for the 480V Cutter Motor and 120V Auger Motor. This system utilizes mechanical contactors (driven by low-voltage MOSFET drivers) to control motor power states. A Variable Frequency Drive (VFD) is integrated into the 480V circuit to control the frequency, and therefore the RPM and torque, of the Cutter Motor.

The Telemetry & Feedback System monitors critical machine parameters. Key metrics include Residual Current Monitoring (RCM), component temperatures, motor RPM, motor torque, and drilling depth.

- **Ground Fault Detection:** An RCM sensor is connected directly to the main 480V feed to continuously monitor for current imbalances, creating an immediate hardware trip in the event of an earth leakage fault.
- **Thermal Monitoring:** NTC Thermistors are installed on motors and gearbox housings to monitor thermal operating limits. These signals are read by the ADC and processed by the Safety MCU.
- **RPM Monitoring:** Rotational speed is monitored via Hall Effect sensors. Neodymium magnets are mounted to the rotating shafts; as they pass the stationary sensor, a digital pulse is generated. The MCU calculates RPM based on the time interval between pulses.

- **Torque & Current:** Motor torque is estimated by monitoring phase current. The Cutter Motor utilizes a Split-Core Current Transformer (CT), while the Auger Motor utilizes a Hall-Effect Current Sensor (ACS712). These values are used for both performance monitoring and jam detection.
- **Voltage Monitoring:** A voltage transducer is installed on the 480V line to monitor voltage sags or instability from the generator source.
- **Depth & Speed:** An Ultrasonic Sensor is mounted on the stationary frame, facing the moving carriage. It emits ultrasonic pulses to measure the distance to the carriage. The software calculates the total depth drilled and the Rate of Penetration (ROP) by comparing these distance changes over time against a set reference point.

Table 5 List of all the sensors on the VBM with their location

Metric	Component	Signal Type	Connected To
Auger RPM	Hall Effect Sensor	Digital Pulse	Teensy 4.1
Cutter RPM	Hall Effect Sensor	Digital Pulse	Teensy 4.1
Analog Conversion	TI ADS1115 ADC	I2C	Teensy 4.1
Auger Current	ACS712 Sensor	Analog (0-5V)	ADC
Cutter Current	Split-Core CT	Analog (0-5V)	ADC
Cutter Voltage	Transducer CR4310	Analog (0-5V)	ADC
Depth/Speed	Ultrasonic Sensor	Digital (UART/PWM)	Teensy 4.1
Auger Control	MOSFET Driver	Digital (3.3V)	Teensy 4.1
Cutter Control	MOSFET Driver	Digital (3.3V)	Teensy 4.1
Ground Fault	RCM Status	Digital (Dry Contact)	Teensy 4.1

Temperature	Thermistor	Analog	ADC
Power Status	UPS Battery Signal	USB / Digital	Raspberry Pi 5
Auger Reverse Control	MOSFET Driver	Digital (3.3V)	MCU
Watchdog Heartbeat	External Watchdog Adafruit TPL5110	Digital Pulse	MCU
Safety Light	Stacklight / Beacon	Digital (3.3V)	MCU
Cutter Aux Contact	Auxiliary Switch	Digital (Input)	MCU
Auger Aux Contact	Auxiliary Switch	Digital (Input)	MCU

3.4.2 Electronic Control Units

The VBM control architecture utilizes a Dual-ECU Design to separate nondeterministic high-level processes from real-time safety-critical control.

ECU 1: Safety Microcontroller (Teensy 4.1) The primary safety controller is a Teensy 4.1 MCU running at 600MHz. This unit is responsible for "Hard Real-Time" tasks, including:

- **Deterministic Loop Execution:** Reading all analog and digital sensors at a fixed 100Hz frequency.
- **Safety Logic:** Executing the primary state machine and managing the hardware watchdog timer "heartbeat."
- **Actuation:** Directly controlling the MOSFET drivers for the motor contactors and reversing logic. The Teensy acts as the ultimate authority on machine safety; it will reject any command from the Host PC if safety interlocks (e.g., temperature, current) are not satisfied.

ECU 2: Host Computer (Raspberry Pi 5) - The Host Computer handles "Soft Real-Time" and user-facing tasks. It hosts the Python/Tkinter Operator GUI, manages the SQLite telemetry database, and handles network communication. It calculates high-level performance metrics (such as Rate of Penetration) and issues high-level state commands (e.g., "Request Start," "Request Reverse") to the Safety MCU.

3.4.3 Components Communication

The communication architecture employs a redundant, dual-link physical layer to ensure data transmission between the Control Logic (Raspberry Pi 5) and the Safety MCU (Teensy 4.1). The Primary Link is established

via a hardwired Ethernet connection. This provides a high-bandwidth, low-latency channel for telemetry streaming and command execution during nominal operation. This link handles the high-frequency (10Hz+) data stream required for real-time graphing and logging. The Redundant Fallback Link utilizes a direct UART Serial connection between the Raspberry Pi's GPIO header and the Teensy's hardware serial pins. In the event of an Ethernet switch failure or network stack error, the system automatically downgrades to this serial link. While bandwidth is reduced, this channel maintains sufficient throughput to transmit critical "Heartbeat" signals, safety status flags, and E-Stop commands, ensuring the system maintains positive control even during a partial network failure.

3.4.4 Power Consumption per Subsystem

Table 6 Power consumption by subsystem breakdown

Subsystem	Voltage	Nominal Load	Peak Load / Inrush	Notes
Cutter Motor	480 VAC 3 ϕ	7.3 kW	10.2 kW	12.3 HP Motor (VFD Limited)
Auger Motor	120 VAC 1 ϕ	0.3 kW	1.6 kW	0.25 HP Motor (Inrush Limited)
Actuation Bus	24 VDC	60 W	100 W	Safety Relay, Contactor Coils, RCM
Control Logic	5 VDC (Batt)	15 W	25 W	Pi 5, Teensy, Sensors
Total System		~7.7 kW	~12.0 kW	Well below 100 kW Limit

3.4.5 Full System Schematic

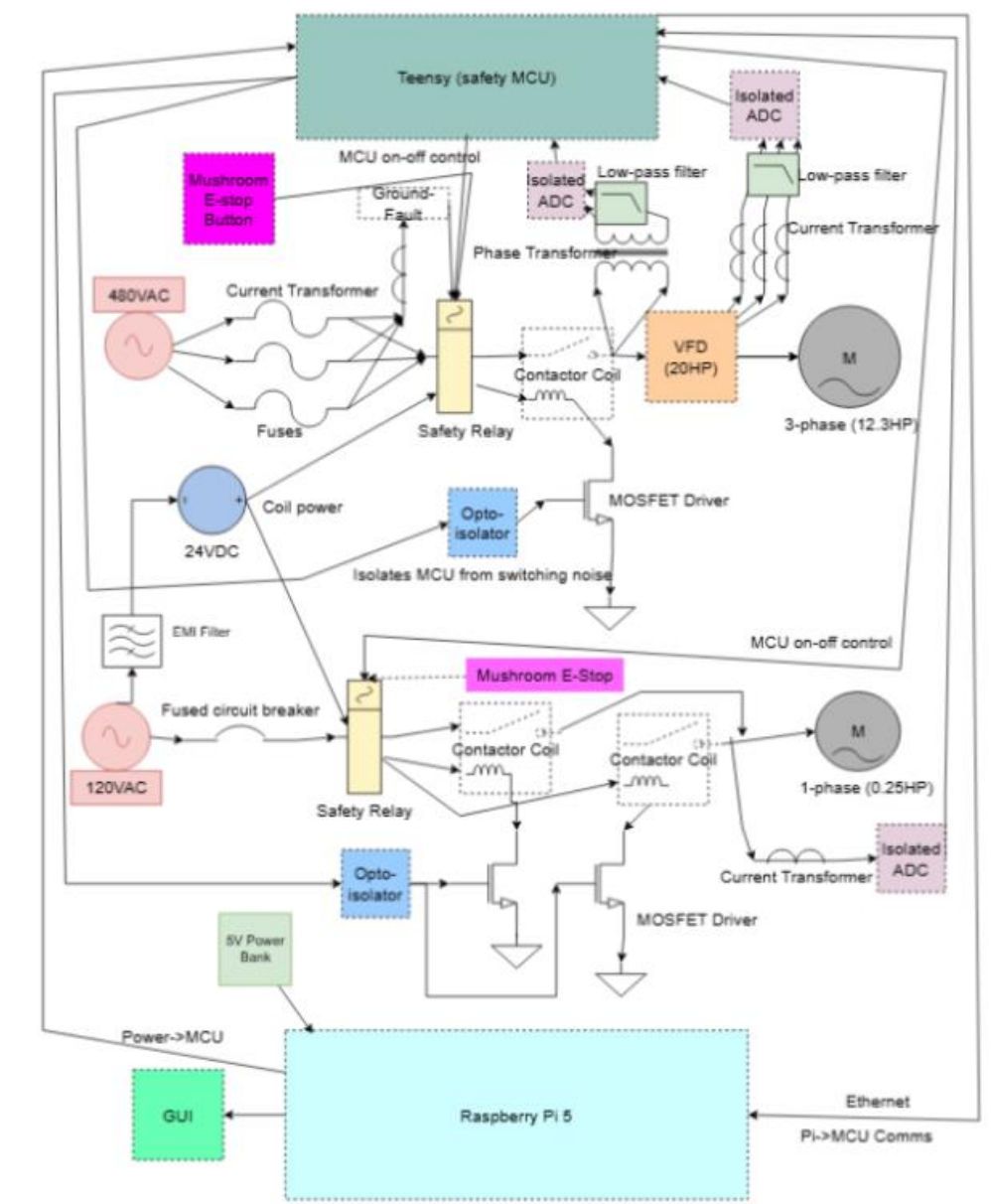


Figure 15 Full electrical system schematic

3.5 Software System

3.5.1 Software System Explanation

The software architecture operates on a hierarchical "Master-Slave" relationship between the Host Computer (Raspberry Pi 5) and the Safety Microcontroller (Teensy 4.1). While the Raspberry Pi manages high-level directives, the Safety MCU firmware executes a finite state machine (FSM) consisting of three primary states: ESTOPPED, IDLE, and RUNNING.

1. ESTOPPED State (Default/Safe): Upon boot or any fault condition, the Safety MCU defaults to this state. In this mode, all MOSFET drivers are de-energized (0V), ensuring contactors remain open. The system monitors the status of the physical E-Stop chain and the "Safety Reset" flag from the Raspberry Pi GUI. The machine cannot transition out of this state until the Two-Action Reset Sequence is completed: the physical button must be unlatched (Hardware True), and the Raspberry Pi must transmit a specific "Reset" command (Software True).

2. IDLE State (Ready): Once the safety chain is verified, the system enters IDLE. The Safety MCU begins toggling the Hardware Watchdog pin to energize the safety relay. It continuously polls the "Heartbeat" signal from the Raspberry Pi. If the Pi crashes or the Ethernet link fails (heartbeat lost for >1.0 seconds), the MCU autonomously reverts to ESTOPPED. In this state, the machine awaits a specific "Start" command from the Pi.

3. RUNNING State (Active): Upon receiving a valid "Start" command from the Raspberry Pi, the MCU closes the relevant contactors and enters the primary control loop. This loop executes parallel monitoring of all subsystems:

- **Power Monitor:** Checks voltage stability and RCM status.
- **Motor Monitor:** Verifies current draw is within limits and detects stall/jam conditions.
- **Performance Monitor:** Compares Commanded vs. Actual RPM. If any sensor violates its safety threshold, the MCU overrides the Pi's commands and triggers a system-wide trip, returning to the ESTOPPED state.

3.5.2. Software Communications

The internal software architecture utilizes a Publish-Subscribe (Pub-Sub) model implemented via the ZeroMQ (ØMQ) messaging library. This decoupling allows the Data Logger, GUI, and Control Logic to operate independent processes that subscribe to the specific data topics they require without tightly coupled dependencies. Data serialization is handled using timestamped JSON (JavaScript Object Notation) packets. This format ensures readability for debugging and provides a consistent schema across all subsystems. A dedicated "Heartbeat" packet is broadcast at 10Hz; if this specific packet is not received by the Safety MCU for a duration exceeding 1.0 seconds, the hardware watchdog logic is triggered to save the machine.

3.5.3. E-Stop Implementation Description

The Emergency Stop system utilizes two IDEC HW1R-V4F10-R mechanical latching switches located in accessible locations on the surface and near the launch frame. These switches are wired in series directly to the Omron G9SE Safety Relay Logic Unit. The system operates on a "De-Energize to Trip" principle: pressing the button physically breaks the 24V control circuit feeding the safety relay. This causes the relay to instantaneously open, physically cutting power to the MOSFET drivers and motor contactor coils, regardless of the software state.

Each E-Stop utilizes a mechanical latching mechanism that locks the button in the depressed (open) position until manually reset by an operator. This prevents automatic re-energization of the machine. The system enforces a strict Two-Action Reset Sequence to return to a ready state:

1. **Hardware Reset:** The operator must physically twist and unlatch the E-Stop button.
2. **Software Reset:** The operator must acknowledge the safety clearance by pressing the "Safety Reset" button on the Operator GUI.

The GUI passively monitors the status of the safety chain via an auxiliary contact but has no authority to override the physical E-Stop. This architecture eliminates the single point of failure present in software-only designs by shifting primary safety authority to the hardware layer.

3.5.4. Machine Operator Controls and Interface

The Operator Interface (GUI) is designed to meet the competition requirement allowing an untrained observer to determine system health within 10 seconds. The interface is divided into four functional zones:

1. **Safety & Control Command:** Located on the left, this zone features a prominent, always-accessible virtual Emergency Stop button. Directly below is the "Safety Reset" button. This button is the software component of the required two-action reset sequence; the machine cannot be re-energized until the physical E-Stop is unlatched *and* this software reset is engaged.
2. **Sensor Health Status:** The center panel utilizes a "Traffic Light" status system for all critical sensors (Current, Voltage, Temperature, RPM).
 - **Green:** Value is within nominal operating limits.
 - **Red:** Value has exceeded safety thresholds, or the sensor is disconnected. This allows for immediate visual identification of specific faults (e.g., "Cutter Motor Temp" turning red) without requiring the operator to interpret raw numerical data during a crisis.
3. **Performance Indicators:** The right-side panel displays calculated process metrics, including Drilling Fluid Pressure, Rotation Torque, and instantaneous Rate of Penetration (ROP), derived from the raw sensor data.
4. **Administrative Control:** A password-protected "Interlock Override" panel is accessible via the top tabs. This feature addresses competition feedback by allowing authorized engineers to temporarily bypass specific non-critical sensor interlocks (e.g., a faulty temperature sensor) to recover the machine, provided safety protocols are maintained.

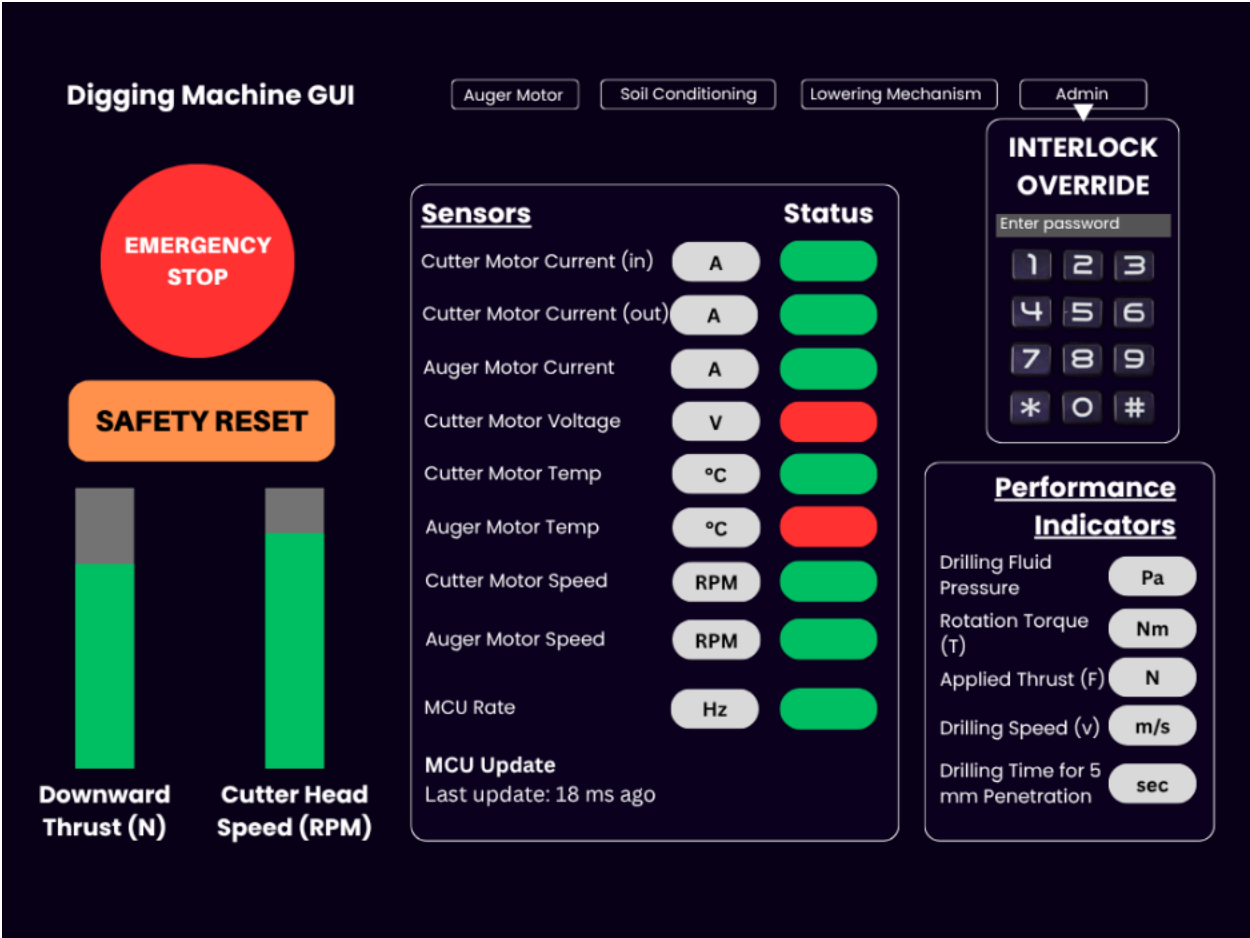


Figure 16 GUI Interface diagram

4.Safety Interlock Mechanism Flow Chart

The Safety Interlock Mechanism Flow Chart illustrates the hierarchical protection strategy, prioritizing hardware-level safety over software control.

Tier 1: Hardware Interlocks The logic begins with a series of physical checks that must be satisfied to energize the Safety Relay. The circuit verifies that the E-Stop Button is closed (unlatched), the Residual Current Monitor (RCM) has not detected a ground fault, and the External Watchdog is active. If any of these hardware conditions are false, the Safety Relay physically opens, triggering a "Hardware Trip" that cuts power to all actuators.

Tier 2: Safety MCU Logic Once the hardware chain is closed, the Safety MCU (Teensy 4.1) performs continuous health checks. It monitors the Heartbeat Signal from the host Raspberry Pi and the status of the Uninterruptible Power Supply (UPS). If a critical fault is detected (e.g., loss of comms or power), the MCU stops toggling the hardware watchdog, forcing a Tier 1 Hardware Trip. Only when all health checks are passed and the Two-Step Reset is performed will the MCU enable the drivers and close the contactors for runtime monitoring.

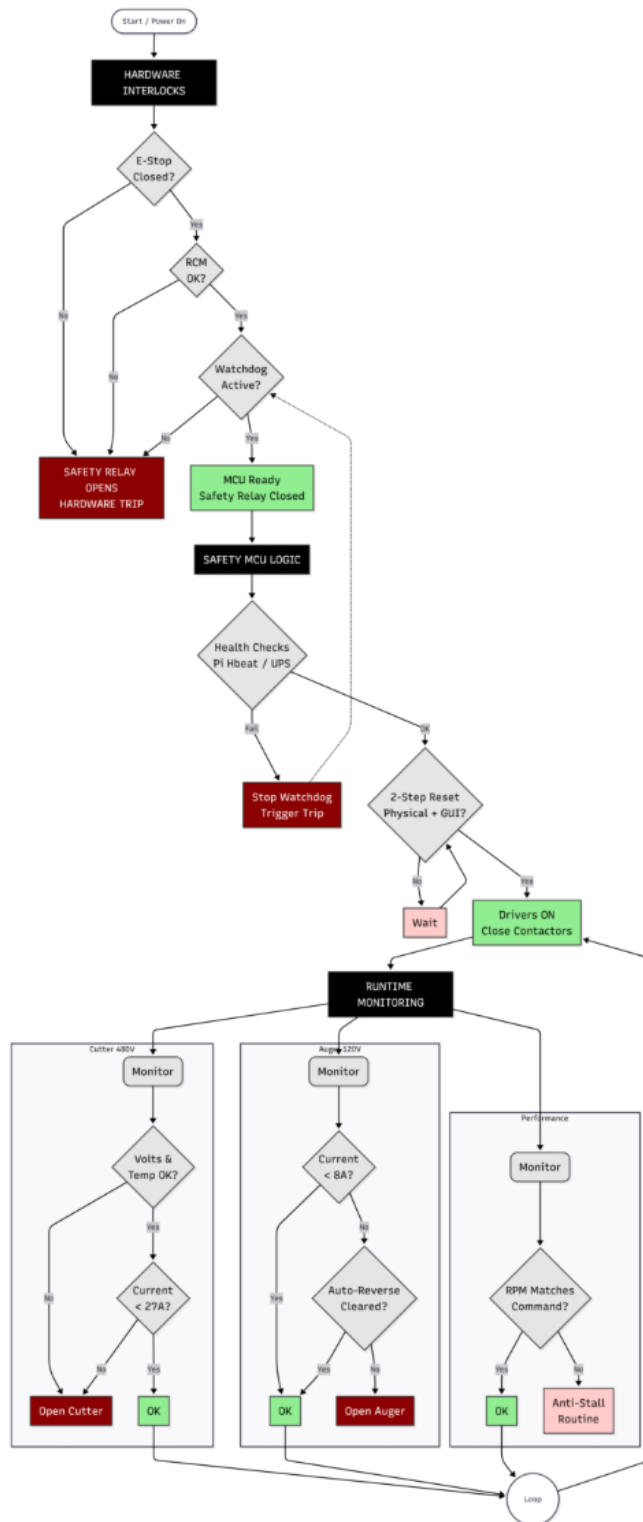


Figure 17 Safety interlock mechanism flow chart

5. Equipment Lifting and Transportation Structural Analysis

5.1 Predicted Loads

The total weight of the machine is estimated to be around 1000lbs. This is a lot of weight, and the team is taking the necessary steps to ensure that we can travel to the competition and move the machine to and from the excavation area. The machine will be transported to the competition in sections to ensure that it is capable of being lifted into the transportation.

5.2 Failure Mode Description

The main failure that the team can predict occurring in terms of lifting and transportation is damage being done to the machine itself. To eliminate this risk the machine will be disassembled and packed very carefully during transportation. Each section will be pinned down during transportation, as well as wrapped in protective material to ensure any banging is avoided. If damage was to be done during transportation, the mitigation would be to either attempt to purchase the necessary materials to repair the damage in Bastrop, or to create a solution there. This would cause more damage to the success of our VBM rather than safety concerns.

In terms of removing the machine from the excavated hole, a handle will be built at the top of the machine to ensure that a forklift is able to latch onto it. Due to the sectional component of the VBM, the team will ensure that each section has a possible latch for the forklift. This is to ensure that if the VBM does come apart while removing it, there are additional opportunities to remove the machine. The team will ensure that all possible power and electrical supplies are turned off during the removal. Damage down to the machine will be mitigated.

5.3 Safety Factors

The main safety factors within the machine itself have been touched on. Throughout the lifting and transport of the machine, the most critical factors are ensuring that every member is careful of any knee, back, shoulder, or other injuries. Transporting the parts there the team will need to ensure that all the parts are secure down, and not in danger of damaging any individuals in the U-Haul. During transport no power supplies will be connected. The handling of any chemicals will be done carefully, always ensuring closure.

5.4 FEA Analysis

The preliminary FEA analysis was completed for the lifting force. This planning is still in the preliminary stages, as the team has been primarily focusing on assembly. However, to ensure that the materials we selected can handle the forces required to lift the machine, a brief FEA analysis was completed. This was completed using a load weight of 800 lb, and carbon steel as the material. As seen in the analysis, the top compartment would be able to withstand the necessary forces since the yield strength far exceeds the maximum load.

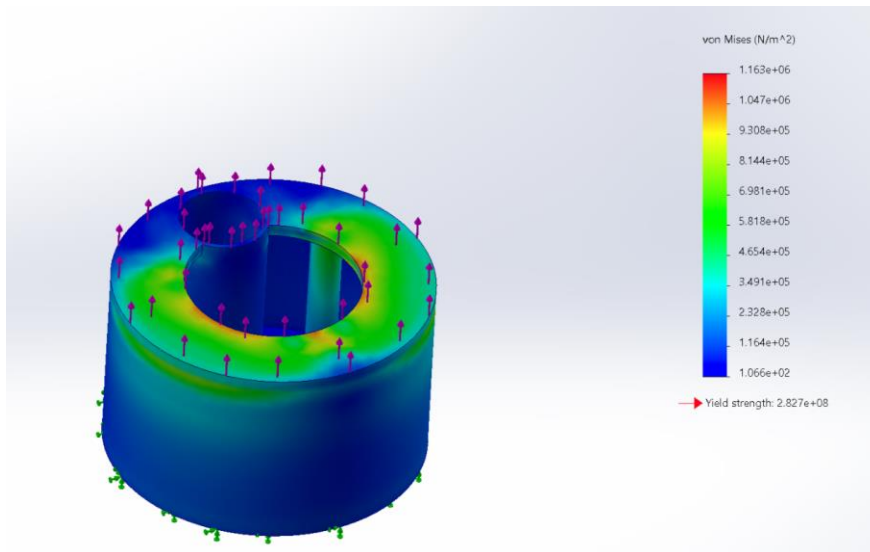


Figure 18 FEA analysis on top of internal structure

5.5 Lifting and Transportation Procedures

Pre Lift

1. Conduct a thorough inspection of the lift area to ensure it is clear of obstacles and hazards.
2. Ensure all members involved in the lift are trained and authorized to perform lifting operations.
3. Members involved must understand the planned lift procedure and goal of the operation.
4. Ensure all members involved in the lift are wearing personal protective equipment (PPE) including gloves and steel-toed boots.

Lift Team

1. Roles are assigned to each member of the lift team, including a lift coordinator, spotters, and operators.
2. The lift coordinator is responsible for overseeing the entire lift operation and ensuring adherence to the lift procedure.
3. Spotters are positioned strategically to monitor the lift and provide guidance to the operators.
4. Operators are trained members who perform the lift.

Lift Procedure

1. Conduct a final safety briefing before initiating the lift, emphasizing key points such as weight limits, hand placement, and emergency procedures.
2. Listen to the lift coordinator for directions during the lift.
3. Lift slowly and steadily, avoiding sudden movements or jerks that could destabilize the load.
4. The lift is not done until the lift coordinator gives an all-clear confirming members have successfully completed the lift.
5. The lift coordinator should be monitoring the lift closely and be prepared to stop and assess if any issues arise during the operation.

6. All lift members should have proper form by bending your knees and lifting with your leg muscles, your spin should not be angled further than 45 degrees.

5.5.1 Road Transportation

The machine itself will be transported to the competition through driving. This is the most realistic choice for the team to evaluate the safety of the machine itself, and the cost. The team will rent a U-Haul or large vehicle to ensure that the machine is able to be transported.

The team is planning to transport the machine while it is “disassembled” as possible, to ensure that the machine can be transported easily. The current design that we have features 3 different main bodies, that will be made individually. Due to the weight of the machine itself,

5.5.2 Securing Machine

The machine will be transported by the team, so it will be critical that additional securing methods are taken. The team will purchase suspension cords and specialized straps to ensure that the machine isn't subjected to any movement during the drive.

5.5.3 First Principles Calculation for Safety Factor

The factor of safety calculation was completed for the lifting force that would be exerted on the top component of the VBM. The factor of safety was calculated to be 247, which shows that it is more than capable of withstanding that force without failure.

$$FOS = \frac{\sigma_{y,vM}}{\sigma_{vM}}$$

$$FOS = \frac{2.87 * 10^8}{1.16 * 10^6}$$

$$FOS = 247$$

6. Functional Test Program

6.1 VBM Testing Procedures

The following section includes all the tests that will be conducted on the VBM.

6.1.1 Rotating of Cutterhead

Cutterhead Performance Test

Objective: Assess the cutterhead's efficiency in drilling through specified soil or rock types.

Procedure:

1. Select representative ground conditions (e.g., clay, sand, or rock).
2. Operate the machine at standard RPM and advance rate.

3. Measure penetration rate (m/min) and cutterhead wear after the operation.

Pass Criteria: The penetration rate must meet the design specifications, and cutterhead wear must not exceed acceptable limits.

6.1.2 Auger Rotation Testing

Auger effectiveness test for removing muck material

Objective: Validate the effectiveness of disturbed clay removal during drilling.

Procedure:

1. Operate the machine while drilling in wet conditions or with significant spoil production.
2. Measure the time and efficiency of the removal process.
3. Inspect for blockages or material buildup on the cutterhead.

Pass Criteria: Removal systems must maintain consistent operation without clogging.

6.1.3 Structural Testing

The internal structure of the VBM can be tested via testing the entire machine and ensuring that the torque does not damage or reach Failure Mode in any internal parts of the machine when the cutterhead encounters resistance. It can also be tested by lifting or adding weight to the machine to ensure that the strain does not damage or impact the

The frame can be tested in a similar way, using applied forces to ensure that all pieces of the HSS are soundly bolted together when the frame is constructed.

Structural Integrity and Vibration Test

Objective: Evaluate the structural soundness and vibration levels during operation.

Procedure:

4. Drill continuously for a specified duration (e.g., 8 hours).
5. Measure vibration levels using accelerometers at critical points.
6. Inspect structural components for signs of stress or damage.

Pass Criteria: Vibration must stay within safe thresholds, and no structural damage should occur.

6.1.4 Control and Software Testing

Test 1: Emergency Stop Hardware Functionality

- **Objective:** Validate the "De-energize to Trip" capability of the physical E-Stop chain.
- **Procedure:**
 1. Energize the machine and enter the "IDLE" state (Safety Relay Closed).
 2. Command the Auger Motor to run at low speed (unloaded).

3. Physically depress the E-Stop button.

- **Pass Criteria:**

- The Safety Relay de-energizes immediately (<20ms).
- The Motor Contactors physically open (audible click).
- The Operator GUI registers an "ESTOPPED" state.
- The machine cannot restart via software until the button is physically unlatched.

Test 2: Watchdog & Communications Loss

- **Objective:** Verify the Hardware Watchdog timer trips the safety relay upon loss of the Host-to-MCU heartbeat.

- **Procedure:**

1. Energize the system to "IDLE" (Watchdog active).
2. Physically disconnect the Ethernet cable connecting the Raspberry Pi to the Teensy.

- **Pass Criteria:**

- The "Heartbeat" signal stops.
- Within 1.0 second, the Hardware Watchdog timer expires.
- The Safety Relay trips, cutting power to all actuators.
- The system defaults to a safe state and requires a full reset sequence to recover.

Test 3: Sensor Limit Interlock

- **Objective:** Confirm the Safety MCU autonomously trips power when a sensor threshold is violated.

- **Procedure:**

1. With the system in "IDLE," apply a simulated fault signal (e.g., 0V or 5V) to the Auger Current Sensor input to simulate a jam (>8A).
2. Attempt to send a "Start" command from the GUI.

- **Pass Criteria:**

- The Safety MCU rejects the start command.
- The GUI displays a specific "Overcurrent" or "Sensor Fault" alarm.
- Actuators remain de-energized.

Test 4: Two-Action Reset Sequence

- **Objective:** Ensure the machine prevents inadvertent restart after a fault.

- **Procedure:**

1. Trigger an E-Stop event.
 2. Unlatch the physical E-Stop button (Action 1) *without* pressing the GUI reset.
 3. Attempt to start motors.
- **Pass Criteria:**
 - The system remains in the "ESTOPPED" state.
 - Motors do not energize.
 - The system only enters "IDLE" after the operator explicitly presses the "Safety Reset" button on the GUI (Action 2).

6.1.5 Sensor and Data Testing

Residual Current Monitor (RCM)

- **Procedure:** With the system powered, press the RCM's built-in "Test" button.
- **Pass Criteria:** Verify that the Safety Relay physically trips (opens) and the GUI display a critical "Ground Fault" alarm immediately.
- **Reset:** Reset the RCM and Safety Relay; verify the system returns to a "Ready" state on the GUI.

Thermistors

- **Ambient Check:** Verify the GUI displays a realistic baseline value ($\pm 5^{\circ}\text{C}$ of room temperature).
- **Response Check:** Apply controlled heat (e.g., heat gun) to the sensor probe and verify the temperature reading on the GUI increases smoothly without signal dropouts or noise spikes.

Hall Effect Sensors (RPM)

- **Pulse Check:** Rotate the drive shaft by hand and verify the GUI registers non-zero RPM pulses.
- **Stability Check:** Run the motor at a low, constant commanded speed. Verify the RPM reading increases from the baseline and remains stable within $\pm 10\%$ of the commanded value.

Motor Current Sensors

- **Zero Check:** Confirm the GUI displays 0 A ($\pm$ noise floor) when the motor is de-energized.
- **Load Check:** Energize the motor and apply a test load. Verify the current value on the GUI increases proportionally with the applied load.

Ultrasonic Sensor (Depth)

- **Static Check:** Position the machine at a known height and verify the depth reading on the GUI matches the physical measurement.
- **Dynamic Check:** Move the machine downward by 50mm. Verify the GUI depth updates accordingly and the "Rate of Penetration" metric registers a positive value.

Analog-to-Digital Converter (ADC)

- **Calibration:** Input a known voltage source (e.g., 1.0 VDC) into an ADC channel. Verify the GUI displays the expected value.
- **Noise Check:** Monitor ADC-driven sensors (thermistors, current sensors) at steady state to ensure signal noise is within acceptable software filtering limits.

Sensor Integration and Data Management

- **Telemetry Stream:** Power on the system and verify all telemetry fields on the GUI are populating with live data.
- **Fault Simulation:** Disconnect a non-critical sensor and confirm the GUI registers a "Telemetry Fault" warning without causing an uncommanded system shutdown.
- **Logging Verification:** Complete a simulated test cycle. Export the SQLite data log and verify that all sensor values were recorded with correct timestamps and formatted correctly for post-run analysis.

7. Machine Production Timeline and Status

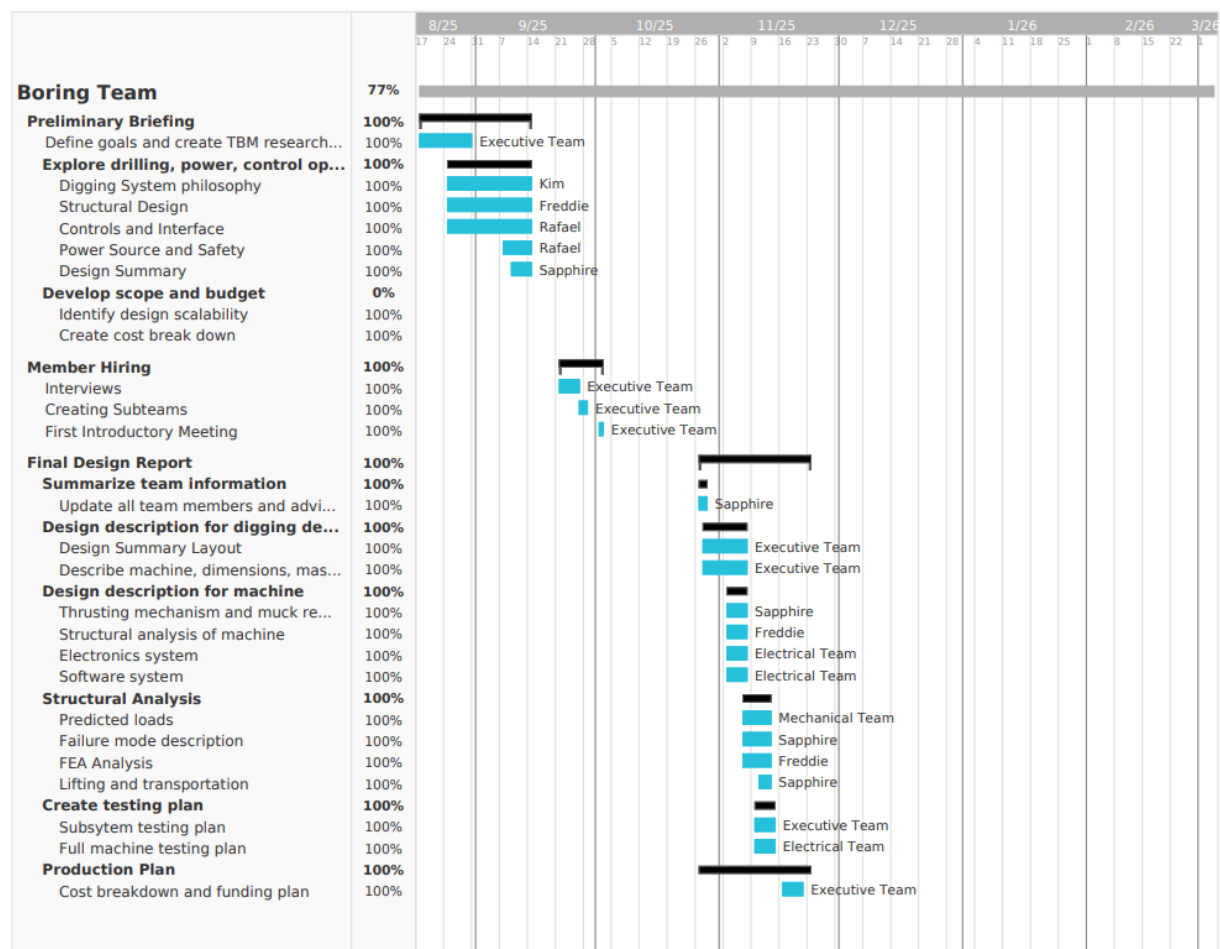


Figure 19 Gantt chart on the preliminary items required for VBM completion

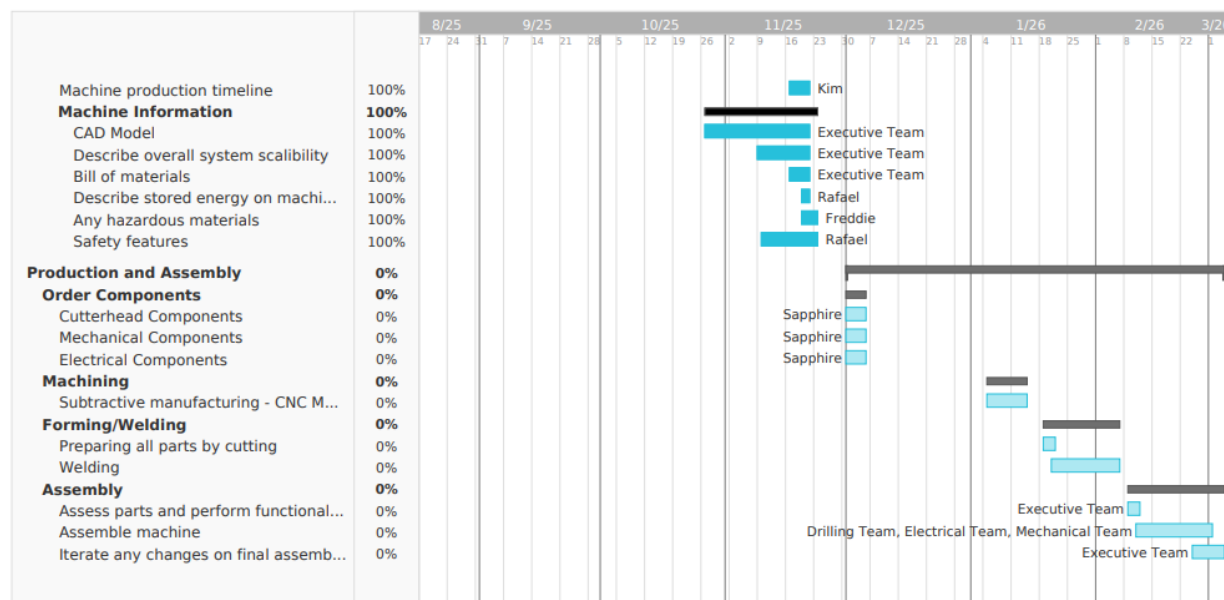


Figure 20 Gantt chart with the remaining items needed to complete the VBM.

8. Machine Cost Breakdown and Funding Plan

The full Bill of Materials with pricing can be seen in 12. CAD 3D Model and Bill of Materials, alongside the pricing. The full price is just under \$16,000. While attempting to lower this price through donations and used parts, we are also aware that our current funding will only cover that. The team is going to attempt to acquire more sponsorship but is also aware that some of the travel costs may have to be out of pocket.

The Boring Team will be funded through sources from Queen's University and corporate sponsors. This includes both the projected machine costs and competition costs of around \$13,000.

The Queen's University funding is projected to be around \$11,000 and will come primarily from the Dean's Donation Fund, Table 11. This fund is available to all Engineering competitive design teams, outreach programs, conferences, clubs and student-led campus businesses which bring distinction to Smith Engineering at Queen's University. Queens provides this financial support to engineering led initiatives to give them opportunities for teamwork, leadership, project planning, design, marketing and communication skills. Teams may apply once a year for funding, the Hyperloop and Boring team applied for this funding in early November and expects to receive the funding in December.

Smith Engineering Hyperloop also has a large business division who are responsible for procuring corporate funding for the team. The team expects a minimum corporate sponsorship of \$2,000 CAD, which will be used primarily to finance the machine costs. The business team plans to take advantage of the number of respected Canadian construction, infrastructure and mining companies which are located and do work in Canada to obtain this funding. The team is currently in funding talks with these companies and will look to secure commitments for funding by January 2025.

Table 7 Funding sources breakdown

Funding Source	Amount	Date Expected
Smith Engineering at Queen's University Dean's Donation Fund	\$11,000 CAD	December 2025
Corporate sponsors	\$5,000 CAD	January 2026

9. Stored Energy

The VBM contains two primary sources of stored energy which are managed according to strict safety protocols:

1. Control System Battery (Primary): The Low Voltage Control Logic is powered by a 20,000 mAh (74 Wh) USB-C Power Bank. This commercially available UL-certified unit allows the data logging and telemetry systems to remain active for up to 8 hours independent of the generator. It is housed in a protected enclosure within the electronics stack.

- *Mitigation:* The battery is physically isolated from the high-voltage motor bus. It powers only 5V/3.3V logic and does not have the capacity to actuate motors.

2. VFD DC Bus Capacitors: The Variable Frequency Drive (Delta VFD220E) contains internal DC-link capacitors that store approximately 0.5–1.0 kJ of energy during operation.

- *Mitigation:* The VFD features internal "bleeder resistors" designed to automatically discharge this energy to safe levels (<50V) within 60 seconds of power removal. As per safety procedures, the team will verify this "Zero Energy State" using a multimeter on the DC bus test points before performing any maintenance, in compliance with Rule 7.g.1.

10. Description of Safety Features

10.1 Mechanisms to Mitigate a Complete Loss of Machine Power

The system is designed to be fail-safe in the event of a complete or partial loss of power. All safety-critical actuation paths (specifically the Omron Safety Relay and the Main Motor Contactors) operate in a Normally Open (NO) state. They require continuous active power to remain energized and closed. Therefore, any loss of power, whether due to an E-Stop activation, cable severance, or generator failure, removes the holding current, causing the spring-loaded contacts to open immediately. This guarantees a controlled stop without requiring software intervention.

10.2 Mechanisms to Mitigate Leaks and Environment Contamination

The team understands that there is a responsibility to ensure that the materials that are being handled adhere to the safety and environmental standards that are set by NABC and are doing everything in their power to eliminate the risk of any leaks or contamination. The main potential hazard that is posed in this area is the possibility of the conditioner leaking into the ground. The material will not contain any toxicity, as it will have been selected to adhere to all the environmental requirements. The team wants to ensure that this material is not wrongly injected into the ground. The primary strategy to ensure this is multi-layer sealing. This will ensure that if one seal fails or is pre-emptively opened that there is a backup to ensure that the liquid stays contained. The machine will be designed to ensure that the container of the liquid is not near anything sharp that could puncture the container and possibly cause a spillage. The machine will be routinely checked to ensure that the fluid dispensing system is in good condition. This subsystem will go through many testing stages before arriving at the competition, so if any additional mechanisms are required, the team will add them before arriving to competition.

10.3 Single Points of Failure Within the Machine

The 2026 system architecture eliminates the single points of failure (SPOF) identified in previous iterations through hardware isolation and redundancy.

Previous designs relied on a Raspberry Pi GPIO signal to control the main relay. If the software froze in a "High" state, the motor could not be stopped. The new design mitigates this via a Hardware Watchdog Circuit (Adafruit TPL5110). This external timer requires a 1Hz "kick" signal from Safety MCU. If the MCU freezes or crashes, the pulse stops, the watchdog timer expires, and it physically trips the Safety Relay logic unit.

The system employs a Dual-Controller Architecture. The user interface is handled by the Raspberry Pi 5, while real-time safety is handled by the independent Teensy 4.1 MCU. If the Raspberry Pi (GUI) crashes, the Teensy detects the loss of the "Heartbeat" signal and executes an autonomous safe shutdown. Conversely, if the Teensy fails, the hardware watchdog trips the power. This ensures that no single processor failure can result in an uncontrolled dangerous state.

10.4 Description of Failure Modes

Table 8 Description on failure modes

Failure Mode	Detection Mechanism	Mitigation / Response
1. Cutter Motor Stall	Current Sensors: Split-core CTs read phase current > 27A (20HP Limit) or RPM Sensors read 0 RPM while commanded to move.	Anti-Stall Logic: The VFD is commanded to stop advancing. The system may attempt to retract slightly to clear the load. If the stall persists >5 seconds, the main contactor opens to prevent motor burnout.
2. Auger Jam	Current Sensor: Hall effect sensor reads >8A on the 120V	Auto-Reverse Routine: The Safety MCU cuts power, waits for the motor to stop, and engages the Reversing Contactor for a 1-second pulse to

	Auger line (indicating a locked rotor).	dislodge the obstruction before attempting forward motion again [Flowchart].
3. Ground Fault (Shock Hazard)	RCM: The Littelfuse Residual Current Monitor detects >30mA leakage to earth on the 480V line.	Hardware Trip: The RCM sends a hard-wired signal directly to the Safety Relay, causing it to open immediately. This is a Tier 1 Hardware Interlock that requires a manual reset.
4. Loss of Comms (Heartbeat)	Safety MCU: The Teensy stops receiving the 1Hz "Heartbeat" packet from the Raspberry Pi (Operator Station).	Watchdog Trip: The Teensy stops toggling the Hardware Watchdog. The Watchdog circuit trips the Safety Relay, returning the machine to a safe, de-energized state until communication is restored.
5. VFD Over-Temperature	Internal VFD Sensor: The Delta VFD reports an over-temp warning via telemetry or digital output.	Cool-Down/Trip: If the temperature exceeds the warning threshold, the control logic reduces the RPM/Load. If it exceeds the fault threshold, the VFD shuts down output to the motor to prevent permanent damage.

Table 9 List of hazards and their mitigations

Hazard	Description of Risk Mitigation	Response
1. General Safety Hazards	General safety hazards include slips, trips, and falls, fall of material, operating dangerous machinery, and electrical hazards. Mechanical members must be trained and supervised while using power tools.	Slips, trips, and falls are to be treated by assessing for head injury, or any sharp pains indicating breaks, strains, or sprains. The individual must seek medical attention for any head related injuries or breaks, strains, and sprains.
2. Electric Shock	Working with electrical systems means there is a risk members could be injured by electrical currents. There are four main types of injuries, electrocution (fatal), electric shock, burns and falls. To mitigate and reduce the chance of this happening the team will ground all electronics by providing a path for fault currents. As well as waterproof enclosures, sealant/coating risky locations.	In the event of an electrical current passing through the body, utilize standard first aid procedure. Shut off all power sources, call 911 if needed.
3. Chemical Usage	Working with soil conditioning chemicals we will have to be aware of any chemical risks that	Members exposed to chemical burns are to seek the MSDS for the

	could arise. Members working with hazardous chemicals may face risk if the chemicals are to contact skin. When working with hazardous chemicals, members are to wear nitrile gloves and lab coats or coveralls, in addition to sealed eye protection. Members are to consult the MSDS prior to beginning work.	exposure procedure. If symptoms of exposure persist, the member is to seek medical attention.
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11. System Scalability

11.1 Scalability of the Cutterhead

The cutterhead frame, teeth, and cutters will likely be reused for the main competition with minimal modifications. The cutterhead diameter can remain the same while still fulfilling the main competition requirements. Models for the rate of excavation and extraction will still be applicable. Depending on the effectiveness of the cutterhead during the mini competition, adjustments may be made to the positioning or number of cutters and teeth.

Since the machine will be operating horizontally, additional stresses applied perpendicularly to the shield and cutterhead frame will need to be considered. This would apply to the components which secure the cutterhead in place, as they would need to be able to handle this change in force.

11.2 Scalability of the Soil Removal System

The auger system can be scaled in the future for a full scale TBM. Augers are used in many full scale TBMs, and while testing may be done on other soil removal systems, data will already be available for an auger system so it will likely be the preferred choice. It will need to be able to handle a higher volume of aggregate, withstand additional wear, and maintain stability over higher pressures. Some challenges with scaling the auger system may include managing larger volumes without blockage, energy needs, and structural demands. The main change in the auger system is the change of angle, as in full scale TBM that auger is at an angle allowing for movement out of the body while the current design has a vertical auger parallel to the body. Scaling strategies will need to entail prototyping, material testing, and a more advanced control system for the new design.

11.3 Scalability of the Power System

The power systems of the tunnel will most likely not be scalable if the team was to compete in the main competition. The motors that have been selected were specifically chosen for the torque, horsepower and RPM that they are able to supply for the downward force. If you were going to create a larger scale machine that needs to exert more power or torque, you would need to replace the motors. However, one of them may be applicable for either the cutterhead or auger. At the very least the motors could be used

11.4 Scalability of the Electrical System

The electrical architecture of the 2026 vertical digging machine was explicitly designed as a modular proof-of-concept for the full-scale horizontal Tunnel Boring Machine required for future competitions. Rather than building a single-use prototype, the team selected industrial-grade components and topologies that can be directly transplanted or expanded for the larger machine.

The core propulsion system utilizes a 12.3 HP Cutter Motor and a Delta VFD220E43A Variable Frequency Drive. As mentioned in Section 11.3, this selection of motor is sufficient for the mini competition but is under-sized to handle the significantly higher torque requirements of horizontal tunneling. However, by validating the VFD's vector control, torque limiting, and regenerative braking parameters now, the team effectively validates the prime mover for the full-scale TBM. The 480V 3-phase distribution backbone (fusing, disconnects, and EMI filtering) is already rated for the higher power demands of a full-scale system and will require minimal modification to support additional drive motors. Second, the Dual-ECU Architecture (Raspberry Pi 5 + Teensy 4.1) is inherently scalable. The communication protocol utilizes a publish-subscribe model (ZeroMQ) over Ethernet. This allows for the seamless addition of new subsystems, such as a guidance ECU, hydraulic controller, or articulation system, without redesigning the core control software. New nodes can simply be plugged into the Ethernet switch and subscribe to the existing telemetry stream. This "plug-and-play" capability significantly reduces the software integration burden for a more complex full-scale machine.

The hard-wired safety chain, anchored by the Omron G9SE Safety Relay, is designed for expansion. The current loop supports the easy addition of further interlocks, such as methane sensors, hydraulic pressure switches, or additional E-Stop points required for a longer tunnel. The "De-Energize to Trip" philosophy and the separation of the 24V safety bus from the logic power ensure that as the system grows in complexity, the fundamental safety guarantee remains intact. Furthermore, the split-power strategy (isolating sensitive 5V logic on a battery/UPS circuit while powering 24V actuation from the mains) solves the critical noise issues common in larger TBMs. Validating this isolation strategy on the vertical prototype proves that the control logic can remain stable even during the high inrush currents and electrical noise generated by a full-scale boring operation. This architecture provides a stable, noise-immune foundation upon which the more complex sensors and navigation systems of the future TBM can be reliably built.

12. CAD 3D Model and Bill of Materials

12.1 CAD Models

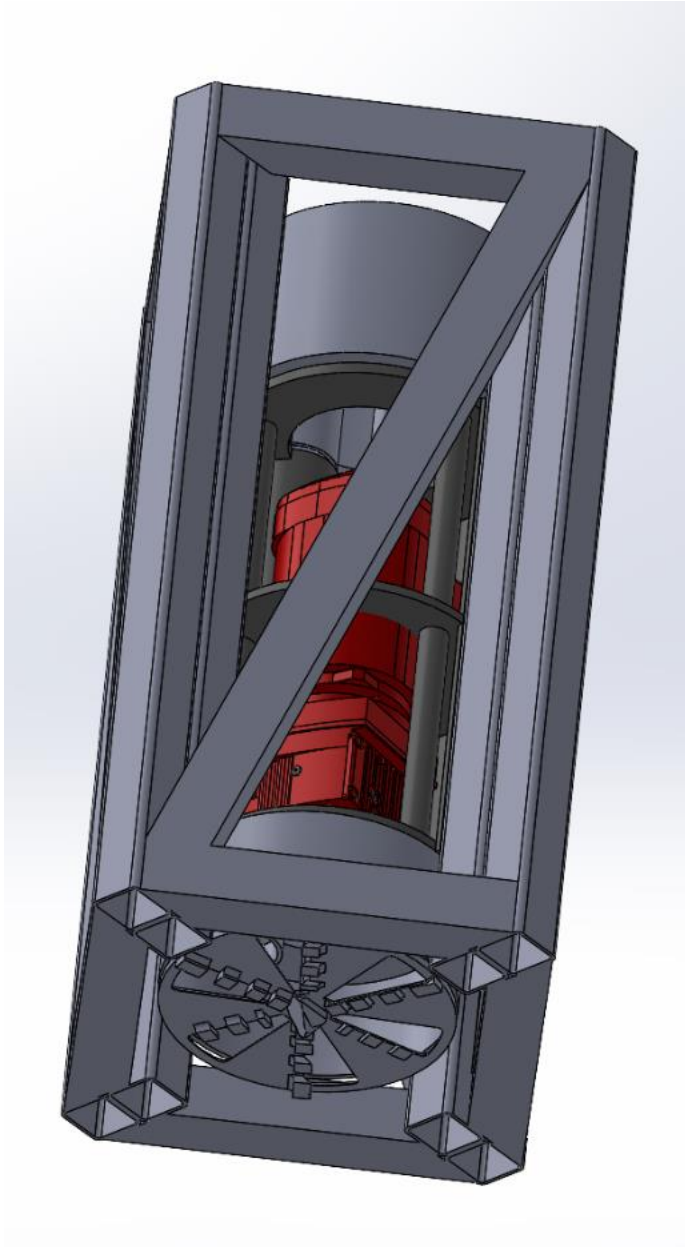


Figure 21 CAD Model, angled view

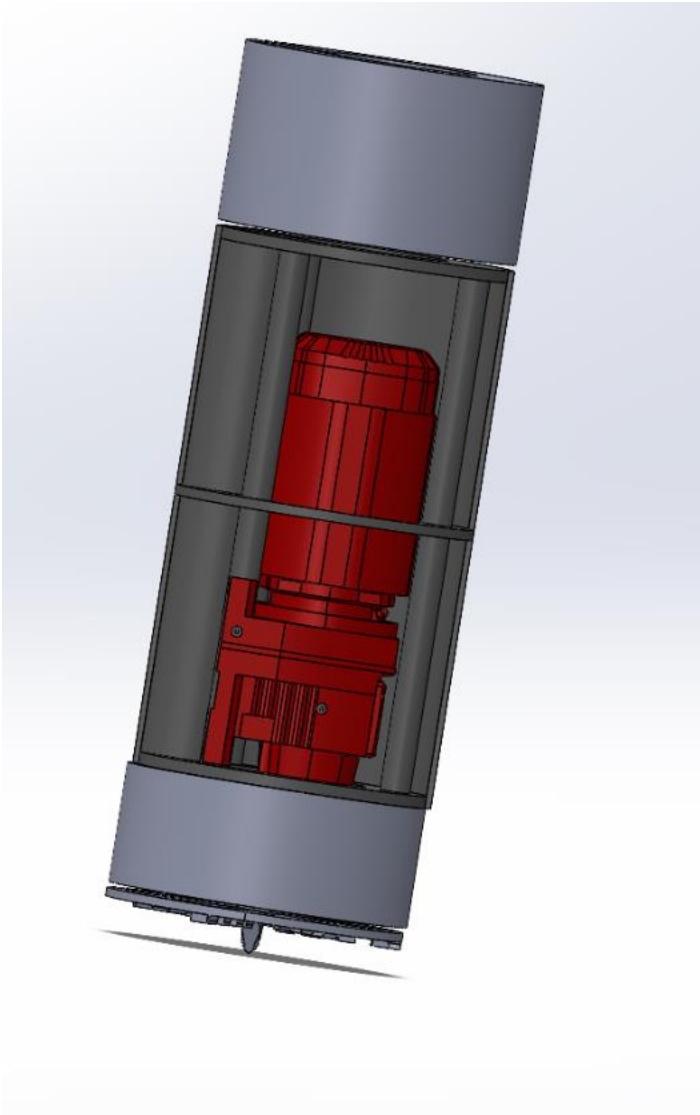


Figure 22 CAD model, side view

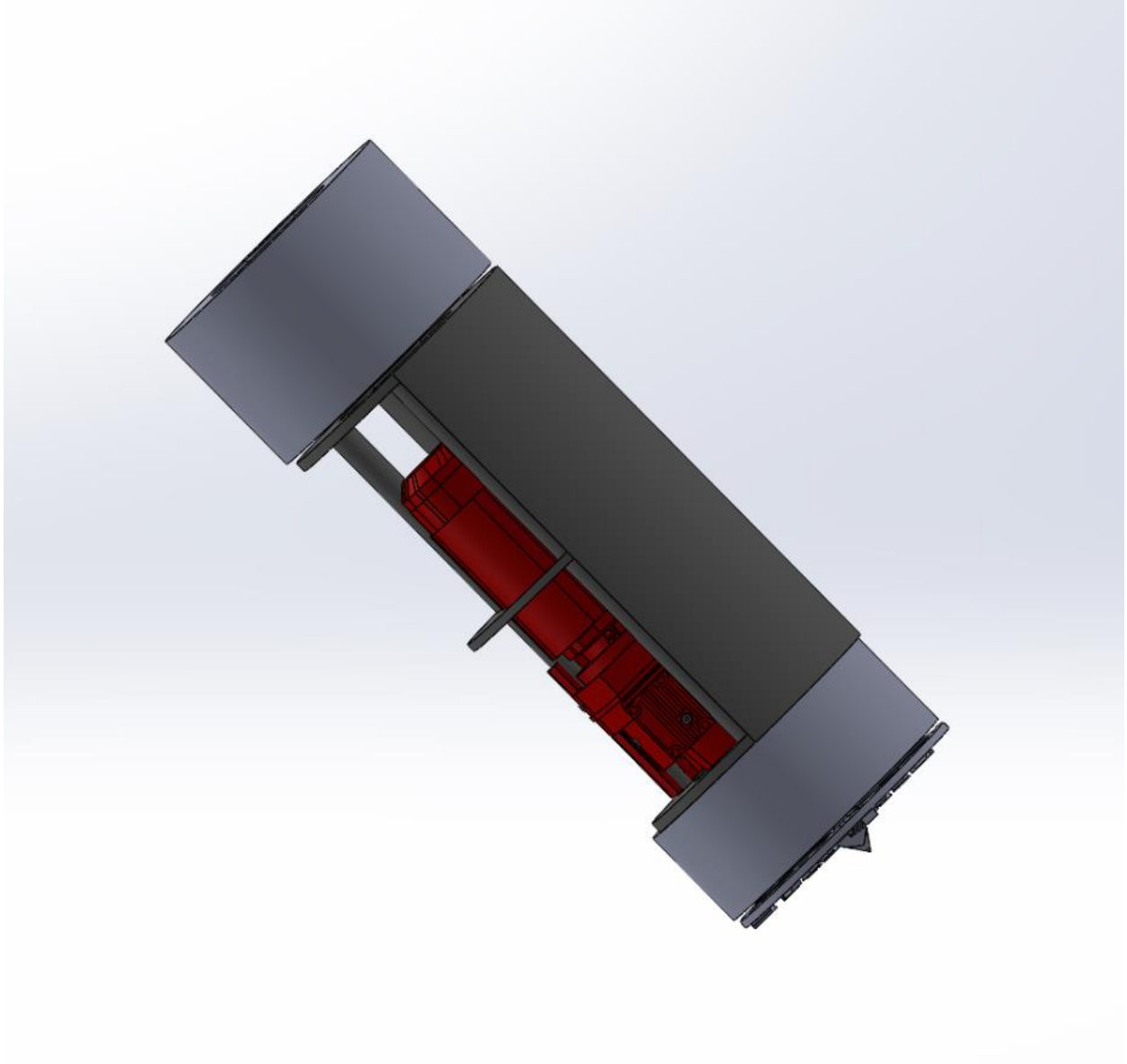


Figure 23 CAD model, additional side view

12.2 Bill of Materials

12.2.1 Electrical BOM

Table 10 Bill of Materials for Electrical parts

Category	Name	Link	Est. Price (CAD)	Qty	Notes
Control & Logic	Teensy 4.1 w/o Ethernet	Mouser	45.07	1	As provided.
	Raspberry Pi 5 (8 GB)	PiShop.ca	111	1	Board only — case & SD extra.
Power Supplies (Low Voltage)	24 VDC PSU (DIN-Rail)	Mouser	65.19	1	100W, 24V/4.2A.
	5 VDC PSU (DIN-Rail)	Mouser	48.65	1	5V/5A — Pi + MCU rail.
	UPS (120 VAC)	Amazon	109.99	1	Powers 24V and 5V PSUs.
Safety System (Hard-Wired)	Safety relay (forced-guided)	Mouser	140.7	2	24V coil.
	E-stop button (latching)	Mouser	31.81	2	22mm push-lock, turn-reset.
480V Power System	VFD (20 HP, 480 VAC)	Digi-Key	1733.91	1	Delta VFD, under budget.
	3-pole Contactor (CRITICAL)	Digi-Key	118.87	1	Correct 38A/20HP w/ 24V coil.
	Safety Contactor (forced-guided)	Mouser	77.3	1	32A, 24V coil.
	3-pole Fused Disconnect	Mouser	265.58	1	60A, 600V, Class J fuses.
	480V Fuses (Class J, 50A)	Mouser	20.2	3	For disconnect.
	Residual Current Monitor (RCM)	Digi-Key	605.15	1	3-phase RCM.
120V Power System	Auger Main Contactor (120V)	Mouser	28.45	1	2-pole, 20A, 24V coil.
	Auger Inrush Current Limiter	Digi-Key	3.04	1	SL22 10008.
	MCB D-curve, 20A	RS-Online	40.88	1	As provided.

Sensors & Peripherals	Split-core CTs	Mouser	56.33	4	0–30A → 0–5V output.
	Water-proof Ultrasonic	Mouser			
	Thermistor temp sensors	Mouser	13.06	4	10k NTC probe.
	24V MOSFET driver module	Mouser	25.86	2	5V logic → 24V/3A output.
	Surge / EMI Filter	Mouser	140.4	1	36A 3-phase input filter.
Budget Categories	Motor branch circuit protection	Digi-Key	904	1	Covers misc breakers/holders.
	Safety sensors & interlocks	Digi-Key	339	1	Limit switches, etc.
	Power & signal wiring	Digi-Key	565	1	8/10 AWG + signal wiring.
	DIN-rail enclosure	Budget	339	1	—
	Labeling, zip ties, misc.	Budget	113	1	—

12.2.2 Mechanical BOM

Table 11 Bill of Materials for Mechanical Parts

Category	Name	Link	Est. Price (CAD)	Qty	Notes
Motors	SEW EURODRIVE R87DRN132L4 Helical Geared Motor	EuroDrive	~ 5000	1	Currently getting quoted
	0.25hp with 170 RPM	EuroDrive	~600	1	haven't quoted
Steel	Carbon Steel Plates	metalsrus.ca	1500.00		Currently getting quoted, x 4 + 1 for Cutterhead
	Internal reinforcing steel	metalsrus.ca	100.00		
HDPE (High Density Polyethylene)	VBM Shell	Iconix	160.00	5-6ft	.5m diameter HDPE

	Auger Shell	Iconix	50.00	.10m diameter HDPE
Gearbox	Gearbox material	metalsrus.ca	300	consult conrad
	Gearbox gears	mcmaster.ca	800	
Shafts	Motor to gearbox shaft	motioncanada.ca	100	<2' length
	Gearbox to auger shaft	motioncanada.ca	50	8" length
Connection Parts	Bolts (around 60)	Home Depot	40	

12.2.3 Drilling BOM

Table 12 Bill of Materials for Drilling parts

Category	Name	Link	Est. Price (CAD)	Qty	Notes
Cutter Teeth	Beaver Style Cutter teeth	Global Tooling	24.28	16	Currently on sale for this price, usually \$32.54
Sealant	Silicone Sealant for Kitchens and Bathroom	Canadian Tire	12.42	1	
Auger	RYOBI 4 in. Earth Auger Bit	Home Depot	70.00	2	
Conditioner	Coordinating with Master Builder Solution	Master Builder Solutions	Donated		